Survival-Supervised Bayesian Matrix Factorization for Prognostic Molecular Programs in Pancreatic Cancer

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Summary

Aim. Bulk pancreatic tumors mix tumor, stromal and immune signals. We ask whether a Bayesian matrix factorization that lets survival inform its molecular programs can find prognostic programs that an unsupervised factorization misses, and whether its risk score transfers to new cohorts.

Method. The model factorizes expression into patient activities and gene loadings and links survival through a Cox model on the projection of each patient's expression onto the loadings. A survival weight w_S sets how much the Cox partial likelihood is weighted relative to the expression likelihood. The primary comparator uses the same factorization at the same rank fitted without survival, followed by a Cox model on its program scores.

Results. On 273 training patients, five-fold cross-validated concordance was 0.663 for the supervised fit and 0.652 for the two-step fit. The external results in this version come from the supervised YFB fit ($w_S = 1$, 2,218 genes, rank 7) and its two-step comparator (Section 1.7). Scored once on 5 external series, their mean concordance was 0.620 against 0.613, a pooled paired difference of $+0.007$ (95% donor-bootstrap interval -0.003 to 0.017). Across the gene sets, the median survival share of the loading precision of the nonempty programs was at most 8×10^{-7}, and the supervised and unsupervised loadings correlate at 1.000. The difference is consistent with joint coefficient estimation under a shrinkage prior. That explanation was not tested separately. Against DeSurv, elastic-net Cox and PurIST, all trained on the same patients, every paired difference in mean external concordance had a 95% interval that included zero (Section 4.8). In simulation, supervision matches the two-step fit when a prognostic program is easy to find. Supervision improved recovery near the detection threshold under strong survival. Choosing the weight by cross-validated partial log-likelihood on the training data kept $w_S = 1$, and the fit keeps all 7 programs because ELBO pruning lost the planted program in simulation.

External cohorts. Every modeling choice is made by simulation and five-fold cross-validation on the 273 TCGA and CPTAC training patients. This covers preprocessing, gene screening, priors, rank, survival weight and comparator tuning. The five external expression series and the matched ICGC methylation set only score a model that is already fixed. Training cross-validation kept the gene screen and the per-cohort standardization and selected rank 5 for the working model (Section 4.6).

Work of November 9–13. The analyses of this week compare priors on the loadings and the survival coefficients, fit methylation alone, test one fitter for several data types, compare two ways of linking survival to the programs, and reselect the gene screen, input scaling and rank on training data. The two prior comparisons and the training cross-validation of the screen, scaling and rank are complete. The current model keeps point-Laplace priors on the scores and loadings and a Normal prior on each survival coefficient (Appendix D). It also keeps the gene screen and the z-scores within cohort, and the one-standard-error rule selects rank 5 (Section 4.6). The other results are in progress or pending review and appear as marked placeholders in Section 3.

1 Methods

1.1 Notation

Patients are indexed by $i = 1, \dots, n$, genes by $j = 1, \dots, p$ and programs (factors) by $k = 1, \dots, K$. Patient i has observed time t_i and event indicator δ_i (1 for death, 0 for censoring). The training data are an $n \times p$ expression matrix Y. For a random quantity θ under the variational posterior q, $\bar{\theta} = E_q \theta$.

Two patient-level quantities are kept apart. The **activities** $L \in \mathbb{R}^{n \times K}$ are posterior scores in the reconstruction $Y \approx LF^\top$ and exist only for training patients. The **program scores** $Z = Y\bar{F}$ are the product of a patient's expression with the posterior-mean loadings. They enter the Cox predictor and can be computed for any new patient whose expression is measured on the same genes.

1.2 Model

The expression matrix is modeled as

$$y_{ij} = \sum_{k=1}^{K} L_{ik} F_{jk} + e_{ij}, \qquad e_{ij} \sim N(0, \tau_j^{-1}), \tag{1}$$

with independent point-Laplace priors on the entries of each column of L and F, a point mass at zero mixed with a Laplace distribution, whose parameters are estimated by empirical Bayes within each program with the R package ebnm (Willwerscheid et al. 2025). Each gene has its own residual precision τ_j. Survival follows a Cox model (Cox 1972) whose linear predictor is the program-score projection,

$$\eta_i = \sum_{k=1}^{K} Z_{ik} \beta_k, \qquad Z = Y\bar{F}, \qquad \beta_k \sim N(0, \sigma_k^2), \tag{2}$$

with the prior variance σ_k^2 estimated per program. This is the YFB model, named for its predictor $\eta = (YF)\beta$. Because new patients are scored by the same projection, prediction needs no refitting.

The variational objective is

$$E_q \log p(Y \mid L, F, \tau) \; + \; w_S \, E_q \log \mathrm{PL}(\beta, F) \; - \; \mathrm{KL}(q \,\|\, \mathrm{prior}), \tag{3}$$

where PL is the Cox partial likelihood. The survival weight w_S multiplies the Cox term wherever it appears: in the loading update, in the coefficient update and in the objective. At $w_S = 1$ the objective is the evidence lower bound of the joint model. Larger values of w_S give survival more influence on the loadings. The full derivation, with every update checked term by term against the code, is in `derivations/SSBMF-YFB-model-derivations/SSBMF-YFB-model-derivation-full.pdf` (Appendix A).

Scale. The reconstruction does not fix the scale of a program: multiplying $L_{\cdot k}$ by c and dividing $F_{\cdot k}$ by c leaves $LF^\top$ unchanged. The fit therefore rescales each nonzero program after a sweep so that its posterior mean training projection has standard deviation 1. Each β_k is then a log hazard ratio per standard deviation of the program score, and a single w_S means the same thing for every program.

1.3 Fitting and prediction

Each sweep re-expands the Cox partial likelihood to second order around the current $\bar{\eta}$, with the predictor's variance taken from $q(\beta)$ and $q(F)$. It then updates, one program at a time, the coefficient, the activities and the loadings. The loading update uses the reconstruction residual and the weighted Cox contribution to its precision and linear term. Each sweep ends by updating the noise precisions and rescaling the programs. The fit starts from the singular value decomposition of Y and stops when the relative change in the objective falls below 10^{-5} or after 300 sweeps. The working fit is unpruned at rank 7 (Section 4.7).

New cohorts are standardized gene by gene within cohort and scored on the training genes they measure. The direction of the risk score is the one the fit learns from training survival. There is no post-fit sign correction: a fit oriented the wrong way would have training concordance below 0.5 and is reported as fitted.

1.4 The two-step comparator

The rank-matched comparison uses the same fitter at $w_S = 0$, which removes survival from the loading update and gives the unsupervised empirical-Bayes factorization at the same rank, on the same genes and patients. Because w_S also multiplies the Cox term in the coefficient update, $w_S = 0$ gives $\beta = 0$. The two-step comparator therefore takes the $w_S = 0$ loadings and fits an unpenalized Cox model on the training program scores. It differs from the supervised fit only in whether survival entered the loadings and in how β was estimated.

1.5 Choosing the survival weight

The real-data fit uses $w_S = 1$, fixed by five-fold training cross-validation (Section 4.1). Two refinements were tested in simulation. In the survival-block form, w_S multiplies the Cox term and the Kullback–Leibler term of β together, so β gets an ordinary Bayesian Cox fit while the loadings see survival at weight w_S. The weight is chosen for each dataset from $\{1, 2, 5, 10, 20, 50\}$ by five-fold cross-validated partial log-likelihood, with ties going to the smaller weight. The same rule was applied to the real training data (Section 4.5).

1.6 Choosing the gene screen, input scaling and rank

The gene screen and the input scaling were compared with alternatives by five-fold training cross-validation on the folds of Section 4.1, using concordance within cohort. The current screen is the 2,218-gene set described in Section 2.1. The alternative is a different preprocessing configuration, not only a different ranking rule. It z-scores each cohort, quantile-normalizes the merged matrix and keeps the 2,000 genes of largest variance. Neither screen uses outcomes, so each was computed once on all training patients, as in DeSurv. The alternative scalings divide each gene by its within-cohort standard deviation, with the Cox projection centered, or apply the within-patient rank transform of DeSurv followed by z-scores within cohort (Young et al. 2026). The current choice was kept unless an alternative was better by more than one standard error of the paired fold difference.

The rank is not chosen by ELBO pruning, which lost the planted program in simulation (Section 4.7). We chose it by five-fold training cross-validation with the one-standard-error rule. The rule takes the smallest K from 2 to 12 whose mean concordance is at least the best mean minus the standard

error of the fold concordances at the best K. Without pruning the fit keeps every column it starts with, so cross-validation chooses the fitted rank.

1.7 The model behind the external results

Every external result in this version comes from the supervised YFB fit ($w_S = 1$, 2,218 genes, rank 7) or its two-step comparator, and each external table and figure names its model in the caption. The supervised fit is `fit_ssbmf_yfb(canonicalize = TRUE, w_S = 1)` on the 2,218-gene set, with every RNA-seq cohort logged once and z-scored gene by gene within cohort, rank 7 without pruning, point-Laplace priors on L and F and a Normal prior on β. The two-step comparator uses the same input and rank at $w_S = 0$, followed by an unpenalized Cox fit (Section 1.4). The survival weight, the gene-set size and the choice not to screen genes on survival were fixed by training cross-validation. The gene screen, the input scaling and the rank were set earlier with external concordance (Section 2.2). Training cross-validation has since kept the screen and the scaling and placed rank 7 on the flat part of its concordance curve, where the one-standard-error rule selects rank 5 for the working model (Section 4.6). If the analyses of Section 3 change the model, the results below keep this label, and the new model is scored once after it is fixed (Section 3.7).

2 Evaluation

2.1 Data

Table 1: Training and validation cohorts. Patient and death counts are read from the saved fold table and external scoring results. Scores omit training genes absent from the external series. PACA-AU is the Australian pancreatic cancer project of the International Cancer Genome Consortium (ICGC). Its array and RNA-seq series share donors. TCGA-PAAD, The Cancer Genome Atlas pancreatic adenocarcinoma cohort. CPTAC, Clinical Proteomic Tumor Analysis Consortium. RNA-seq, RNA sequencing.

Cohort	Platform	Patients	Deaths	Missing genes	Role
TCGA-PAAD	RNA-seq	144	75	0	Training
CPTAC	RNA-seq	129	64	0	Training
Dijk	RNA-seq	90	81	30	External validation
Moffitt	Microarray	123	83	71	External validation
PACA-AU array	Microarray	63	38	13	External validation
PACA-AU RNA-seq	RNA-seq	52	31	27	External validation
Puleo	Microarray	288	181	82	External validation

The model is trained on 273 patients (139 deaths) from two RNA-seq cohorts, TCGA-PAAD and CPTAC (Table 1). Every RNA-seq cohort is transformed once by $\log_2(x + 1)$. Genes are screened separately in each training cohort by the combined mean and variance rank of DeSurv (Young et al. 2026), keeping the top 3,000 per cohort. The screen uses no outcomes. The intersection, called the 2,218-gene set, has 2,218 genes. Each cohort is then standardized gene by gene to mean 0 and standard deviation 1.

The 5 external series supply 616 records and 414 recorded deaths. The PACA-AU array and RNA-seq series share 47 donors, and the RNA-seq series has 52 records from

51 donors, so the external collection represents 568 people if the other cohorts are distinct (counts from the October 16 accuracy review, `paper/writeup/2026-10-16-writeup/_process/Accuracy_Review_Dispositions_10_16_26.md`).

2.2 Uses of the external cohorts

Every modeling choice is made by simulation and by five-fold cross-validation on the 273 TCGA and CPTAC training patients. This includes preprocessing, gene screening, the priors, the rank, the survival weight and the tuning of comparators. The five external expression series (Dijk, Moffitt, PACA-AU array, PACA-AU RNA-seq and Puleo) and the matched ICGC methylation set only score a model after it is fixed. Their results are reported, but they never choose between models or settings. The DeSurv comparator used its published settings rather than a new search on our folds. The gene screen, the per-cohort standardization without a rank transform and rank 7 of the current model were set before this rule, using external concordance. They have since been redone by training cross-validation (Section 1.6, Section 4.6). The audit in `docs/findings/2026-11-06-external-cohort-uses.md` records which earlier choices used which series.

The rule follows the protocol DeSurv used on the same cohorts (Young et al. 2026). DeSurv was trained on TCGA and CPTAC pooled. Every hyperparameter was chosen by five-fold cross-validation on the training data, with the one-standard-error rule for the rank, and the external cohorts were scored once with the final model. No portion of the training data is held out here either. With 273 patients, a held-out fifth would estimate concordance with a standard error near 0.05, larger than the differences between the settings being compared.

2.3 Simulation designs

All three simulation studies plant a single prognostic program among unsupervised structure and ask whether the fit recovers it. Recovery requires absolute loading correlation of at least 0.7, an active coefficient and the correct coefficient sign after loading alignment. A **false-active** program is one whose coefficient lies more than 1.96 posterior standard deviations from zero but which does not match the planted program.

- **The fixed-weight simulation.** Survival weight at 2,000 genes. Data with 50 or 2,000 genes and a planted program explaining 0.25%, 1.00%, 5.00% of expression variance. We fitted 10 datasets per cell at $w_S \in \{0, 1, 10, 100, 1{,}000, 10{,}000\}$, at $w_S = p/50$ and at a weight calibrated to give survival 20% of the loading precision. The generator uses strong planted survival effects.
- **The cross-validated-weight simulation.** The larger gene-count cells of the fixed-weight simulation plus a cell with weaker survival. The weight was chosen per dataset by the rule of Section 1.5. Two fits differ in the coefficient posterior. The block-weighted fit uses a mean-field posterior, with the survival weight multiplying the Cox term and the Kullback–Leibler term of β. The joint-posterior fit is block-weighted with a joint Gaussian posterior for β.
- **The near-threshold simulation.** Programs near the detection threshold. Planted programs explained 0.05%, 0.10%, 0.20% of expression variance, with 10 datasets per cell. The mean oracle concordance ranges were 0.94 to 0.96 for strong survival and 0.73 to 0.74 for realistic survival.

The cross-validated-weight and near-threshold simulations compare the supervised and matched

two-step fits on the same datasets. The unsupervised reference in the fixed-weight simulation was uncentered, so the matched two-step comparison on its cells comes from the cross-validated-weight simulation.

2.4 Performance metrics and uncertainty

Discrimination is Harrell's concordance index C (Harrell et al. 1982), with the risk direction fixed from training. Paired differences between two models use percentile bootstrap intervals (Efron and Tibshirani 1993) from resampling patients within each cohort, with both models scored on the same resamples. The pooled difference is the equal-weight mean of the within-series differences. Calibration is the slope from a Cox regression of held-out survival on the fitted score. A slope above one indicates fitted risk differences that are too small. A slope below one indicates differences that are too large. In simulation the slopes use exact event times, because the default tie handling of `survival::coxph()` merges the extreme simulated times and flattens every slope.

The record bootstrap resamples records within series. The donor sensitivity resamples each donor together with every observed record, so the 47 donors shared by the two PACA-AU series move together. Both pooled intervals use the same equal-weight mean of the within-series differences.

3 Work of November 9–13

This section lists the analyses planned for the week of November 9. The two prior comparisons and the training cross-validation of the gene screen, input scaling and rank are complete, and their results are given below. The other analyses are in progress or pending review, so their subsections give the question and a marked placeholder for the result. Unless a subsection says otherwise, the simulation cells are the three planted variance shares of the fixed-weight simulation (0.25%, 1% and 5% at 2,000 genes) and the near-threshold cell with a 0.05% share and strong survival. The real-data comparisons use the five training folds of Section 4.1, balanced by cohort and event. All fits keep rank 7 and the unit survival weight unless the subsection varies them. None of these analyses uses an external cohort to make a choice.

3.1 Priors on the scores and loadings

A nonnegative factorization, as in DeSurv, makes each program an additive set of genes. This comparison asks whether point-exponential priors on L and F recover a planted prognostic program as well as the working point-Laplace fit, with false activity and convergence under control, and whether they predict as well under training cross-validation. The point-exponential fit takes nonnegative input, each gene divided by its within-cohort standard deviation without centering, and centers the program scores that enter the Cox model. The point-Laplace fit takes the working z-scored input. Both arms are fitted to the pooled training data without cohort strata. Within-cohort concordance is the primary cross-validation metric, with pooled concordance reported beside it.

Result. The point-Laplace prior on z-scores within cohort is kept. In simulation the point-exponential prior recovered the planted program at least as often. It also marked more programs survival-active that matched no planted program, but these carried small coefficients relative to the planted program's. The difference appeared in training cross-validation, where its within-cohort concordance was 0.573 to 0.635 across four input setups, against 0.658 for the current model, and

no setup converged in more than 1 of 5 folds. An independent review on November 12 confirmed these comparisons. Tables and details are in Appendix D.

3.2 Prior on the survival coefficients

Under the L and F prior chosen above, this comparison asks which prior on β the working model should use: point-Laplace, point-normal or Normal. The working fit has used a Normal prior without a comparison that chose it. The three arms differ only in the β prior and use the same simulation cells and training folds. For the point-normal prior the posterior inclusion probability of each program is also recorded.

Result. The Normal prior on each coefficient is kept. A shared point-Laplace, a shared point-normal and a shared Normal prior gave nearly the same fits, with within-cohort cross-validated concordance from 0.649 to 0.658 across the four priors and convergence in the same 3 of 5 folds. Under the shared point-normal prior the posterior inclusion probabilities of the full training fit were 0.94 to 1.00 for all seven programs, so they do not single out the 3 survival-active programs (Appendix D).

3.3 Methylation fitted alone

This analysis asks whether the single-modality model, fitted to methylation alone, finds prognostic programs that the expression fit does not. It uses the matched TCGA patients with expression, methylation and survival. CpGs with mean methylated fraction between 0.1 and 0.9 on these patients (182,934 CpGs) are ranked by variance, and the fit is run on the top 10,000, 25,000, 50,000 and 100,000 and on all of them. Both input and prior setups of Section 3.1 are fitted, with a Normal prior on β. The comparison is with an expression fit on the same patients and folds. A methylation program counts as not seen in expression if its patient scores correlate below 0.5 with every expression program score and it is survival-active in the full fit and in at least three of five folds. With about 150 patients the concordance comparison is descriptive. A full fit at each count is timed first, and cross-validation runs only for counts that converge. If the comparison of β priors adopts a sparse prior, the methylation fits are repeated under it.

> **Note**
>
> **Result pending.** [RESULT C1: convergence and run time at each CpG count for each setup.]
> [RESULT C2: cross-validated concordance of methylation alone against expression alone on the same patients and folds.]
> [RESULT C3: number of methylation programs not seen in expression, with their descriptions.]
> [RESULT C4: scores of the fixed methylation and expression models in the matched ICGC methylation set, reported as description only.]

3.4 One fitter for several data types

This work asks whether one fitter, with shared scores L and a separate loading matrix and prior for each data type, can fit expression alone, methylation alone or both, and so replace the single-modality fitter. The first check is that its single-data-type path reproduces the single-modality fitter in $E[L]$, $E[F]$, $E[\beta]$ and the final objective to a relative tolerance of 10^{-6}. Its Kullback–Leibler helpers are then checked against two properties of the single-modality objective: the Gaussian log-normalizer in

the activity KL is counted once per observation, and the coefficient KL is exactly zero for a flat coefficient likelihood. A two-data-type simulation plants one shared prognostic program, and the real-data step fits expression, methylation and both on the matched TCGA patients with the folds of Section 3.3.

> **Note**
>
> **Result pending.** [RESULT D1: agreement of the single-data-type path with the single-modality fitter on the two test fixtures, as the largest relative difference in each quantity.]
> [RESULT D2: outcome of the Kullback–Leibler helper checks.]
> [RESULT D3: recovery of the shared prognostic program, false-active count and convergence in the two-data-type simulation, and whether methylation dominates the shared scores.]
> [RESULT D4: cross-validated concordance for expression alone, methylation alone and both, and the programs that appear only in the joint fit.]

3.5 Survival through the scores or through the projection

The working model links survival to the projection of the data on the loadings, $\eta = (YF)\beta$, and scores new patients by that projection. The alternative links survival to the latent scores, $\eta = L\beta$, and scores new patients by fold-in, which solves for their scores under the stored priors. This comparison fits both on identical input, priors, starting values and folds, so only the parameterization of the hazard differs. It reports simulation recovery, training cross-validation, the programs of the two full fits and, as description only, the external scores of the two fixed models.

> **Note**
>
> **Result pending.** [RESULT E1: recovery, false-active count, convergence and validation concordance of the two parameterizations in each simulation cell.]
> [RESULT E2: fold-wise and mean cross-validated concordance of the two fits and their paired difference with its standard error.]
> [RESULT E3: comparison of the programs of the two full training fits, including top genes, overlap and the correlation between L_k and $(YF)_k$.]
> [RESULT E4: external concordance of the two fixed models, labeled with their model.]

3.6 Training cross-validation of the gene screen, input scaling and rank

Three choices of the current model were set with external concordance (Section 2.2). Each is redone here by five-fold training cross-validation, as described in Section 1.6. Two outcome-free gene screens are compared: the current 2,218-gene screen and a variance screen after quantile normalization. The input scalings compared are z-scores within cohort, division by the within-cohort standard deviation, and a within-patient rank transform followed by z-scores. The rank is selected over $K = 2, \dots, 12$ by the one-standard-error rule, with the number of programs with nonzero loadings and of survival-active programs reported at each K.

Result. The current screen and the z-scores within cohort are kept, and the rank changes from 7 to 5. No alternative screen or scaling was better than the current input in within-cohort concordance. Concordance was flat from $K = 5$ to 12 and lower below $K = 5$, and the one-standard-error rule

selected $K = 5$ (Section 4.6). The external results in this version come from the $K = 7$ fit, which lies on the flat part of the curve. The working model now uses $K = 5$, adopted from one run of five-fold cross-validation. At that rank the point-exponential prior on expression divided by its within-cohort standard deviation still loses to the point-Laplace prior (Appendix D). An independent review on November 12, 2026, confirmed these comparisons.

3.7 External scoring of the fixed working model

If the comparisons above change the working model, the external results in the rest of this version stay labeled with the supervised YFB fit ($w_S = 1$, 2,218 genes, rank 7). The new model is fixed on training data first and then scored once on the external cohorts.

> **Note**
>
> **Result pending.** [RESULT EXT: external concordance, calibration slopes and pooled comparisons of the working model as fixed after this week's analyses, with its model; or a statement that the working model did not change.]

4 Results

4.1 Supervised fit against the rank-7 two-step fit

Table 2: Five-fold training cross-validation on the 2,218-gene set (single log). Both fits have rank 7 and use the same folds. The two-step fit takes the loadings of the $w_S = 0$ fit and fits a Cox model on the training program scores. Survival-active: coefficient at least 1.96 posterior standard deviations from zero in the fit to all training patients (Wald test for the two-step fit). Convergence refers to the factorization. C is Harrell's concordance index. SE is standard error across folds.

Fit	Mean fold C (SE)	Pooled out-of-fold C	CV partial log-likelihood	Folds converged	Survival-active
Supervised, w_S = 1	0.663 (0.024)	0.663	−775.2	3 of 5	3
Two-step (w_S = 0 loadings, then Cox)	0.652 (0.026)	0.647	−777.1	3 of 5	4

On the training patients, the supervised fit had mean fold concordance 0.663 against 0.652 for the two-step fit (Table 2). The paired fold difference was +0.011 (SE 0.005). In 2 of the five folds the supervised fit reached the 300-sweep cap with the objective still changing slowly. Their coefficients were small, and no coefficient or noise precision was growing, so these folds showed no sign of divergence.

This configuration ($w_S = 1$, the 2,218-gene set, no survival screen) was fixed from these training results before any external cohort was scored. It was then scored once on the five external series (Table 3). This is the supervised YFB fit ($w_S = 1$, 2,218 genes, rank 7) of Section 1.7. Every external result in this section comes from it and its two-step comparator.

Table 3: External concordance of the fixed supervised fit and the two-step fit. Each series was scored once, after the configuration was fixed by training cross-validation. C intervals are 95% percentile bootstrap intervals over patients. The difference is supervised minus two-step on the same resamples. The pooled comparison reports intervals from resampling records and resampling donors, plus a sensitivity excluding PACA-AU RNA-seq. The calibration slope is the coefficient of a Cox regression of held-out survival on the fitted risk score. A slope of 1 means the training coefficients predict hazard differences of the observed size. Model: supervised YFB fit (w_S = 1, 2,218 genes, rank 7) and its two-step comparator.

Series	C, supervised (95% CI)	C, two-step	Difference (95% CI)	Slope, supervised	Slope, two-step
Dijk	0.584 (0.513 to 0.658)	0.567	+0.018 (0.000 to 0.035)	0.55	0.35
Moffitt	0.537 (0.459 to 0.611)	0.531	+0.007 (−0.014 to 0.026)	0.36	0.28
PACA-AU array	0.656 (0.562 to 0.751)	0.656	0.000 (−0.026 to 0.025)	0.78	0.73
PACA-AU RNA-seq	0.681 (0.579 to 0.780)	0.671	+0.010 (−0.019 to 0.038)	1.57	1.26
Puleo	0.644 (0.599 to 0.686)	0.643	+0.001 (−0.007 to 0.010)	0.99	0.80
Mean of five series	0.620	0.613			
Pooled, record bootstrap			+0.007 (−0.002 to 0.017)		
Pooled, donor bootstrap			+0.007 (−0.003 to 0.017)		
Pooled, without PACA-AU RNA-seq			+0.006 (−0.003 to 0.016)		

The supervised fit scored equal to or higher than the two-step fit in every series. Its mean external concordance was 0.620 and the two-step fit's 0.613. The pooled paired difference was +0.007 (95% interval −0.002 to 0.017), and +0.006 (−0.003 to 0.016) without the PACA-AU RNA-seq series. The donor bootstrap interval is −0.003 to 0.017 for the full panel. All pooled intervals include zero. The difference is about the size of the training cross-validation difference. Dijk and Moffitt were the weakest series for both fits.

The supervised slope was closer to one in 4 of 5 series. In PACA-AU RNA-seq it was 1.57, compared with 1.26 for two-step. In Puleo it was 0.99.

As a summary on the scale DeSurv reports (Young et al. 2026), the hazard ratio per standard deviation of the risk score, from a Cox model stratified by series over all 616 records, was 1.37 (95% CI 1.24 to 1.51) for the supervised fit and 1.35 (1.22 to 1.49) for the two-step fit.

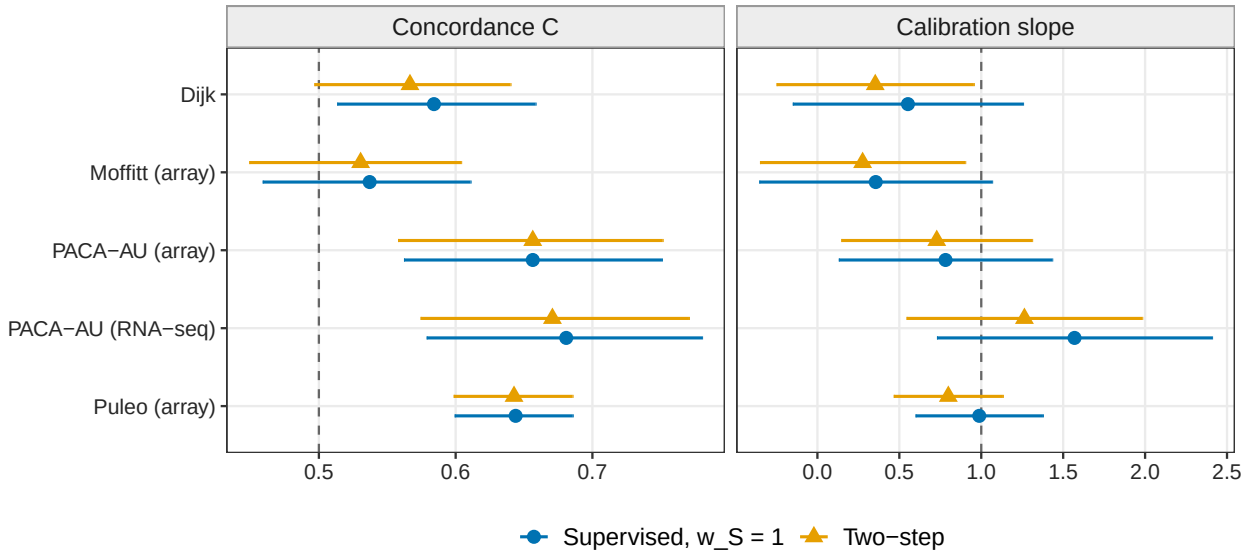


Figure 1: External concordance with bootstrap intervals and calibration slopes with Wald intervals for the two fixed fits. Model: supervised YFB fit (w_S = 1, 2,218 genes, rank 7) and its two-step comparator.

4.2 What survival does to the real-data loadings

On the training data at $w_S = 1$, survival supplies between 2×10^{-10} and 8×10^{-7} of the loading precision (median over programs, across the gene-set sizes below). The simulation cell medians at the same weight ranged from 0.01 to 0.20 at the larger gene count. Two things keep it small on real data. The fitted coefficients are small (largest $|\bar{\beta}_k|$ 0.41 per standard deviation of the program score), and the share scales with $E[\beta_k^2]$. The real programs also explain much more expression variance than the planted programs in simulation, so their expression precision is large. The loading correlations between the fits round to 1.000 for every matched pair. The concordance difference in Section 4.1 reflects the joint coefficient update with its shrinkage prior.

Table 4: Training cross-validation by input gene-set size. Sets are nested and outcome-free, ranked by the per-cohort mean and variance rank. The 2,218-gene set is the default. Match: absolute loading correlations of the two survival-active programs of the all-genes supervised fit with their best matches at each size.

Genes	Supervised C (SE)	Two-step C (SE)	Folds converged, supervised	Match
2,218	0.663 (0.024)	0.652 (0.026)	3	0.78 0.89
3,000	0.663 (0.026)	0.653 (0.027)	3	0.72 0.80
5,000	0.657 (0.030)	0.652 (0.031)	4	0.69 0.77
8,001	0.647 (0.030)	0.650 (0.033)	5	0.74 0.97
12,001	0.643 (0.037)	0.653 (0.037)	5	0.96 0.80
17,900	0.648 (0.038)	0.652 (0.037)	5	1.00 1.00

Adding genes beyond the 2,218-gene set did not help (Table 4). Supervised concordance ranged from 0.643 to 0.663 across sizes, and every size was within one standard error of the others. The size was chosen as the smallest set within one standard error of the all-genes concordance whose

survival-active programs match the all-genes fit at $|r| \geq 0.7$. Training cross-validation identified the 2,218-gene set as optimal for both fits. The supervised advantage over the two-step fit shrank as genes were added and reversed at larger gene-set sizes.

Screening genes on survival inside each training fold, by a cohort-stratified univariable Cox test, gave no consistent gain over an outcome-free screen of the same size: the paired differences were 0.001 to 0.021 across sizes and fits, each within about one paired standard error, and no screened set beat the supervised fit on all 2,218 genes. With survival permuted in simulated data, the $w_S = 1$ fit had 0.1 to 0.5 false-active programs per dataset on average.

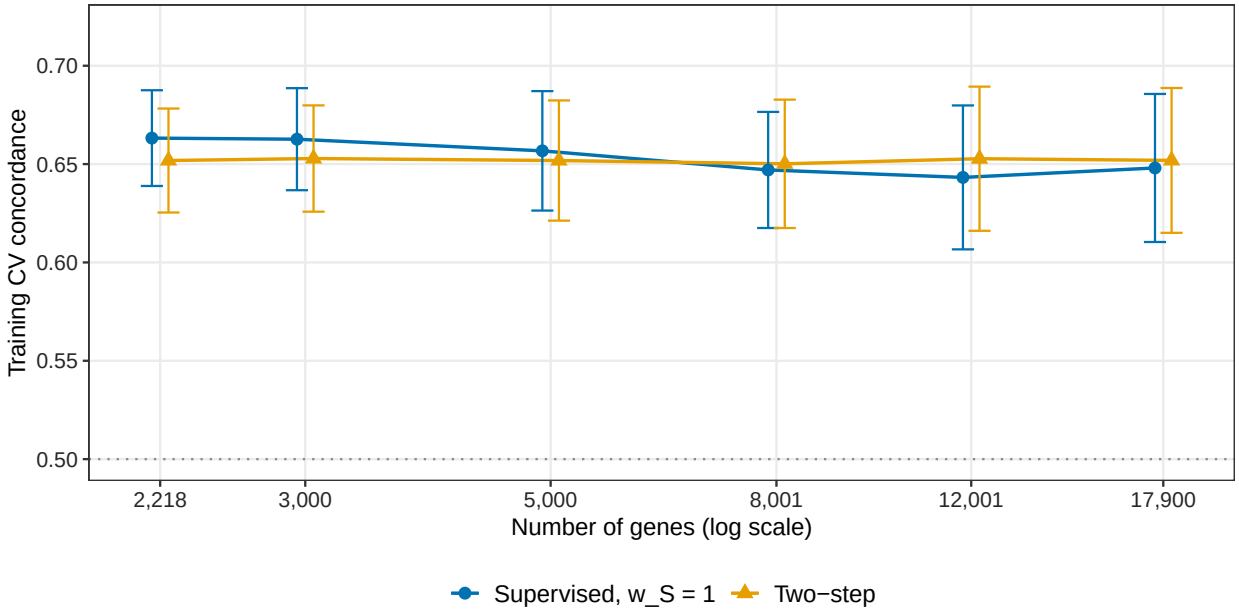


Figure 2: Cross-validated training concordance by gene-set size. Points give fold means with one standard error.

4.3 Simulation: supervision strength and the weight choice

At $w_S = 1$ with 2,000 genes the fit recovered the planted program in all 10 datasets per variance cell, with mean false-active counts 0.9, 0.1, 0.0, and every fit converged. No fit at a weight of 40 or greater converged at the larger gene count. Training CV selected the unit weight in 55 of 60 datasets. The coefficients at $w_S = 1$ were attenuated: median validation calibration slopes were 1.45, 2.16, 3.23 across the three cells, where one denotes calibration. Mean validation concordance ranged from 0.944 to 0.986.

Table 5: Weight chosen by five-fold cross-validated partial log-likelihood (the cross-validated-weight simulation), against the two-step fit on the same datasets. The share cells are the 2,000-gene cells of the fixed-weight simulation, labelled by planted variance share. The weaker-survival cell uses a smaller coefficient. Ten datasets per cell. Paired entries show supervised / two-step. False-active is the mean false-active count. C is mean validation concordance. Slope is the median calibration slope with exact event times.

Fit	Cell	Weights (count)	Recovered	False-active	C	Slope
Block-weighted	0.25% share	1:1 2:9	10 / 10	0.9 / 1.7	0.946 / 0.939	1.24 / 1.16
Joint-posterior	0.25% share	2:10	10 / 10	0.6 / 1.7	0.946 / 0.939	1.22 / 1.16
Block-weighted	1% share	5:10	10 / 10	0.0 / 0.9	0.972 / 0.972	1.27 / 1.10
Joint-posterior	1% share	5:10	10 / 10	0.0 / 0.9	0.972 / 0.972	1.28 / 1.10
Block-weighted	5% share	10:10	10 / 10	0.0 / 0.2	0.986 / 0.986	1.45 / 0.96
Joint-posterior	5% share	10:10	10 / 10	0.0 / 0.2	0.986 / 0.986	1.45 / 0.96
Block-weighted	Weaker survival	1:8 2:2	10 / 10	1.2 / 0.5	0.671 / 0.668	1.05 / 1.04
Joint-posterior	Weaker survival	1:10	10 / 10	0.9 / 0.5	0.672 / 0.668	1.05 / 1.04

In the cross-validated-weight simulation the weights selected by partial log-likelihood increased with the planted signal (Table 5). Both supervised arms and the matched two-step fit recovered the planted program in every dataset. Mean validation concordance differed by at most 0.007. Each arm had a median calibration slope between 0.8 and 1.25 in 1 of 3 variance cells, so the coefficients stayed attenuated. The mean-field arm's largest mean null false-active count at its modal weight was 1.1. The mean concordance of the joint and mean-field coefficient variants differed by at most 0.001.

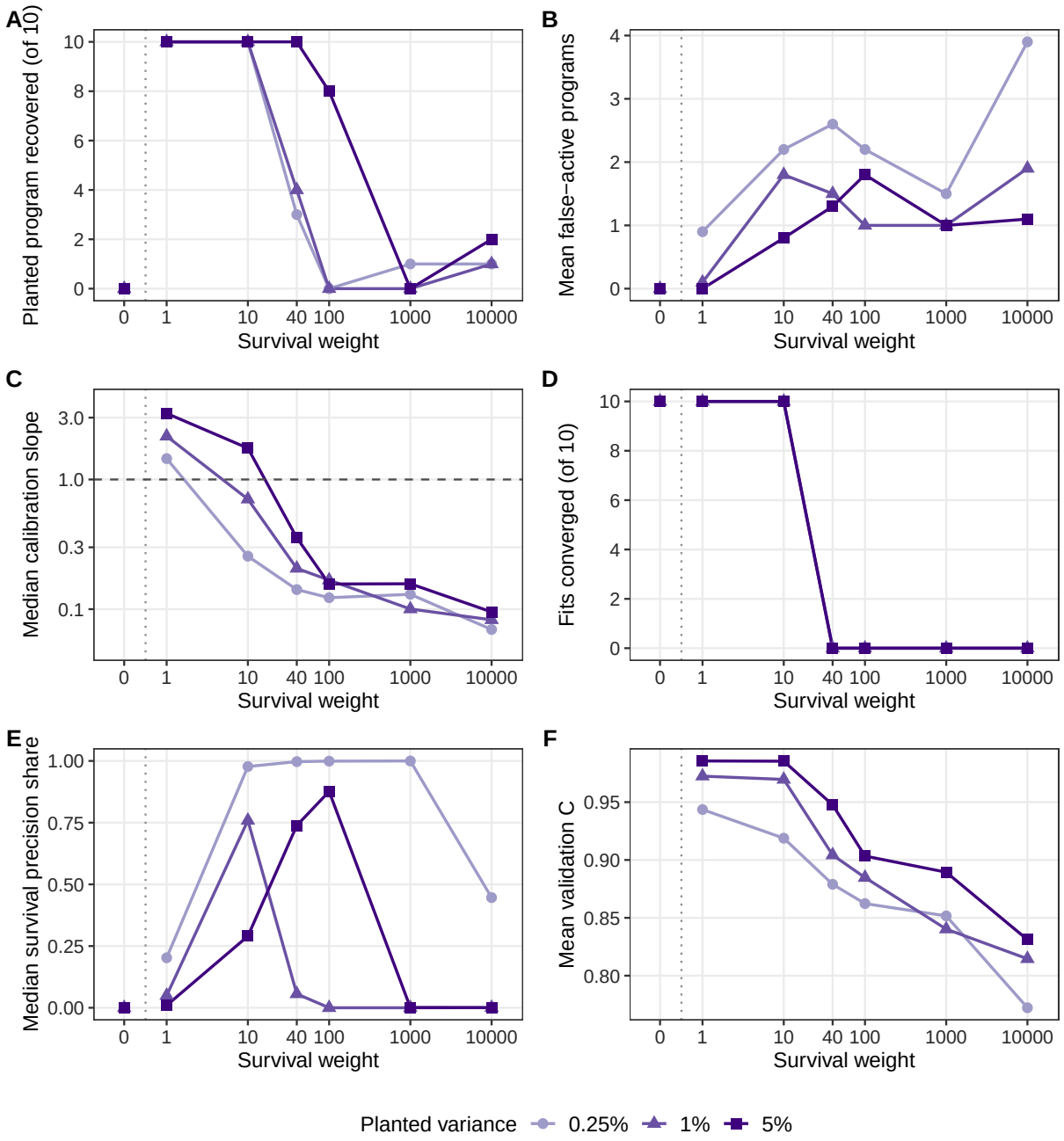


Figure 3: The fixed-weight simulation at the larger gene count. The panels show recovery, false activity, exact-time calibration, convergence, loading precision share and validation concordance.

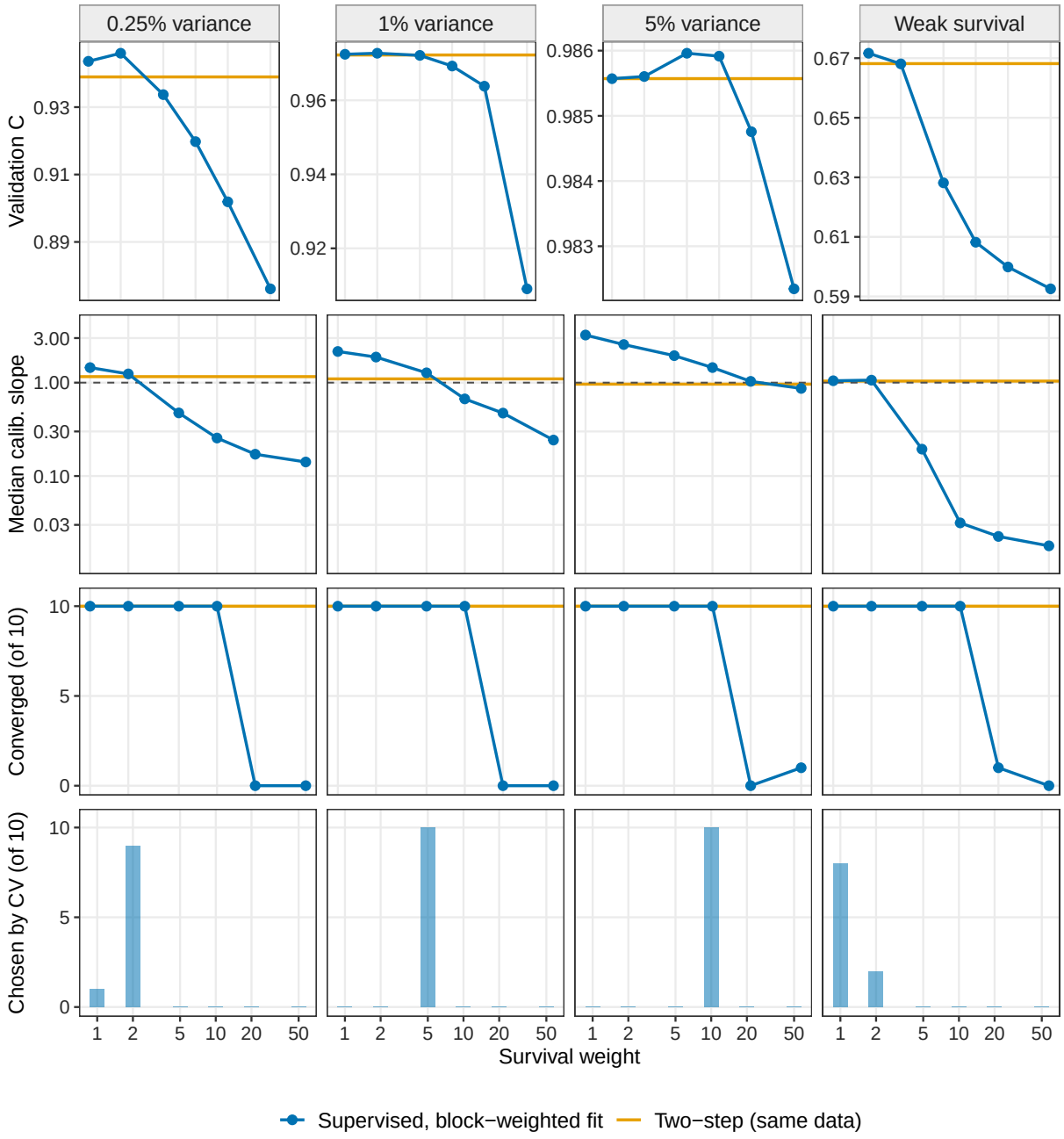


Figure 4: Cross-validated-weight simulation: validation concordance, calibration and convergence by weight, with the matched two-step reference and the CV selection frequencies. The concordance range varies by cell.

4.4 Simulation: programs near the detection threshold

Table 6: Programs near the detection threshold (the near-threshold simulation). All fits converged. Planted share is the planted program's share of expression variance. Entries show supervised / two-step for recovery, false-active count and C. Difference is the mean paired validation C of the supervised fit at w_S = 1 minus the two-step fit, with the number of datasets in which the supervised fit was higher. Oracle C is the validation concordance of the true predictor.

Survival	Planted share	Recovered	False-active	C	Difference (wins)	Oracle C
realistic	0.05%	0 / 0	1.9 / 1.6	0.630 / 0.592	+0.038 (9)	0.740
realistic	0.10%	9 / 8	1.0 / 1.1	0.712 / 0.705	+0.008 (8)	0.736
realistic	0.20%	10 / 10	0.5 / 0.8	0.728 / 0.727	+0.002 (8)	0.735
strong	0.05%	5 / 0	2.5 / 2.8	0.800 / 0.644	+0.156 (10)	0.936
strong	0.10%	10 / 8	1.6 / 2.4	0.901 / 0.861	+0.041 (10)	0.948
strong	0.20%	10 / 10	1.0 / 1.8	0.936 / 0.928	+0.008 (9)	0.958

In the cells of the fixed-weight and cross-validated-weight simulations the two-step fit already finds the planted program, so those simulations cannot show what supervision adds. The near-threshold simulation planted weaker programs (Table 6). With strong survival the supervised fit had the higher validation concordance in 29 of 30 datasets, and the gain grew as the program weakened. At the smallest planted share, 0.05%, the supervised fit recovered the program in 5 of 10 datasets, where the two-step fit recovered none, and validation concordance rose by +0.156. With realistic survival the two fits were close at the larger shares, and neither recovered the program at the smallest share. Both fits averaged more than one false-active program per dataset in every strong cell, the two-step fit as many or more than the supervised fit. Their contribution to the risk score is measured next.

The post hoc analysis measures the variance of the sum of the nonmatched score contributions relative to the variance of the full score. It includes nonmatched programs below the survival-activity threshold. It also measures full-minus-matched concordance on the same datasets. In the cross-validated-weight simulation the largest cell median variance ratio was 0.8%, and the largest absolute cell mean change in concordance was 0.002. In the larger-share cells of the near-threshold simulation, the supervised ($w_S = 1$, block-weighted) and two-step fits the corresponding ratios reached 1.3% for supervision and 2.9% for two-step. Across both fits the largest absolute mean change in C in those cells was 0.008. At the smallest planted share the median ratios ranged from 8.7% to 68.8%.

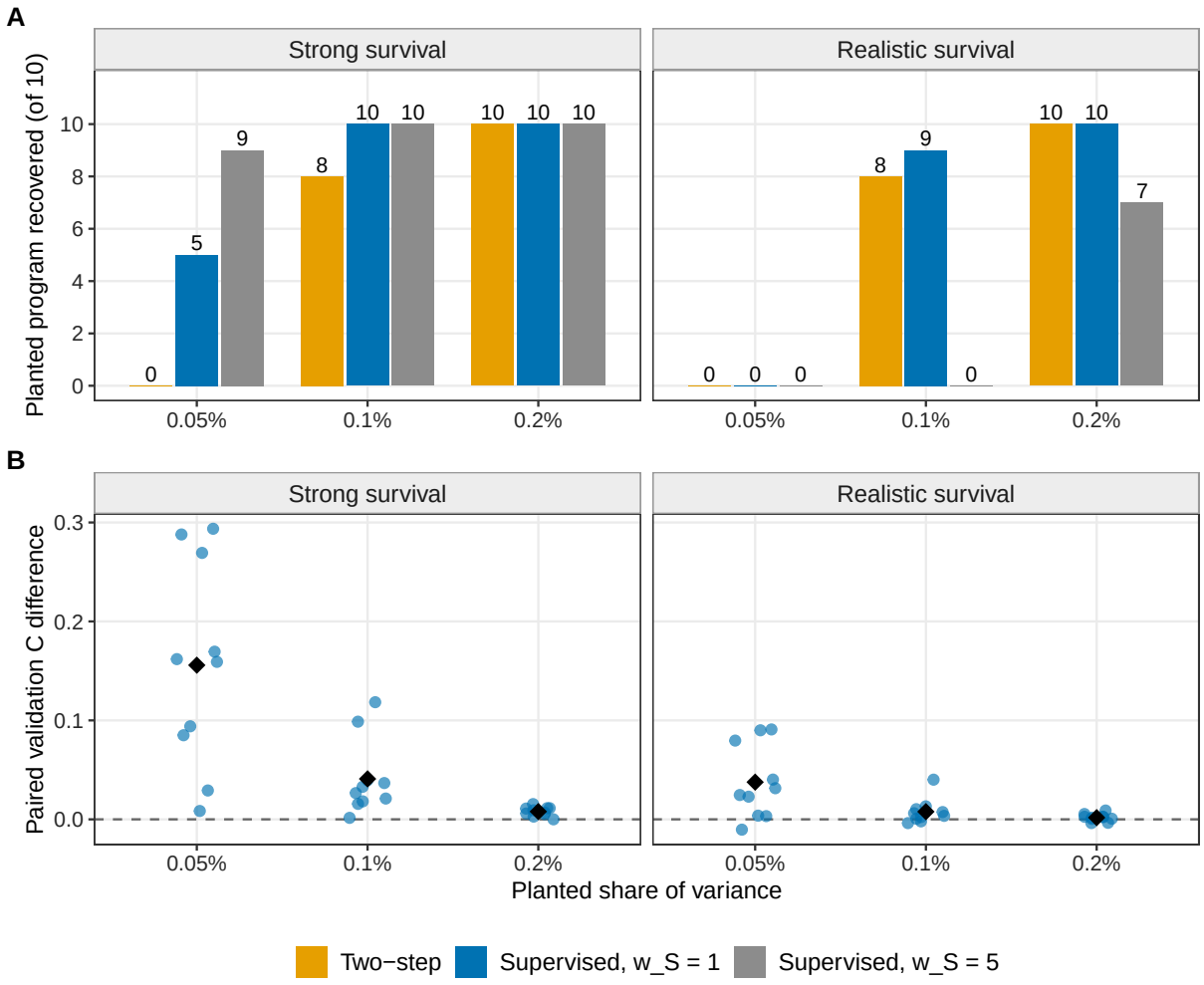


Figure 5: Near-threshold simulation: recovery by fit and paired validation concordance differences for the unit-weight fit versus two-step. Dots in the difference panels represent datasets. Diamonds give cell means.

4.5 Real-data refit at the cross-validated weight

The largest cross-validated partial log-likelihood on the training folds occurred at $w_S = 1$. Across the tested grid from 1 to 50, the cross-validated partial log-likelihood varied by 0.26. The nested folds selected weights 5, 2, 50, 1, 10. Nested cross-validation gave mean C 0.656 (SE 0.027), compared with 0.663 for the $w_S = 1$ fit. At the largest tested weight the median survival share for active loadings was 7.3×10^{-5}. Cross-validation kept $w_S = 1$, so the external results of Section 4.1 apply unchanged.

Table 7: Training cross-validation of the survival-block weight. The precision share is the median among active programs. The nested evaluation uses weights chosen within each outer training set.

Weight	CV partial log-likelihood	Mean C (SE)	Folds converged	Survival precision share
1	−775.168	0.663 (0.024)	3/5	1.17e-06

Table 7: Training cross-validation of the survival-block weight. The precision share is the median among active programs. The nested evaluation uses weights chosen within each outer training set.

Weight	CV partial log-likelihood	Mean C (SE)	Folds converged	Survival precision share
2	−775.179	0.663 (0.025)	3/5	2.34e-06
5	−775.213	0.661 (0.025)	4/5	5.91e-06
10	−775.244	0.661 (0.025)	4/5	1.20e-05
20	−775.380	0.659 (0.026)	4/5	2.49e-05
50	−775.429	0.653 (0.028)	4/5	7.26e-05

4.6 Gene screen, input scaling and rank in training cross-validation

Training cross-validation supported the gene screen and the input scaling of the reported fit (Table 8). In concordance within cohort, the alternative screen was lower by 0.012 (paired SE 0.014), division by the within-cohort standard deviation by 0.020 (SE 0.007) and the rank transform by 0.017 (SE 0.009). No alternative was better, so the current screen and the z-scores within cohort stay. Each alternative converged in more folds than the current input.

Across ranks, concordance was flat from $K = 5$ to 12, between 0.649 and 0.666, and lower from $K = 2$ to 4, between 0.611 and 0.620 (Table 9). The best mean was at $K = 10$ (0.666, SE 0.027), so the threshold was 0.639 and the one-standard-error rule selected $K = 5$. Its concordance was 0.003 below that of $K = 7$ (paired SE 0.007). Every fold fit at $K = 5$ converged, against 3 of 5 at $K = 7$, and the full fit had 3 survival-active programs at every rank from 5 to 8. Every column kept nonzero loadings at every rank, so the priors did not remove programs. The paired differences are much more precise than the fold standard error used by the rule, and on that scale ranks 5 to 8 cannot be told apart. The rule picks the smallest rank on the plateau. It does not show that $K = 5$ predicts better than $K = 7$. The external results in this version come from the $K = 7$ fit, and the working model now uses $K = 5$. Its five programs match Programs 1 to 5 of the $K = 7$ fit, with score correlations of 0.92 to 0.995, and the two programs dropped were not survival-active. Programs 2 and 5 stay survival-active. Programs 3 and 4 lie near the 1.96 threshold in both fits: Program 3 falls below it at $K = 5$ (z from 3.0 to 1.5) and Program 4 rises above it (z from 1.9 to 2.4). The program classification will be repeated at $K = 5$. An independent review on November 12, 2026, confirmed these comparisons.

Table 8: Training cross-validation of the gene screen and input scaling at rank 7 (five folds). Concordance counts only pairs of patients from the same cohort, so the reported fit's value differs from the mean fold concordance in Table 2. Differences are paired over folds. Convergence is within 300 sweeps. Independently reviewed on November 12, 2026.

Input	Genes	Mean C (SE)	Difference from reported fit (SE)	Folds converged
Current screen, z-scores within cohort (reported fit)	2,218	0.658 (0.026)		3/5
Variance screen after quantile normalization, z-scores within cohort	2,000	0.646 (0.030)	−0.012 (0.014)	5/5
Current screen, divided by within-cohort SD	2,218	0.638 (0.023)	−0.020 (0.007)	5/5
Current screen, within-patient ranks then z-scores	2,218	0.641 (0.022)	−0.017 (0.009)	5/5

Table 9: Training cross-validation of the rank on the current screen with z-scores within cohort (five folds). Concordance counts only pairs within cohort. Differences from K = 7 are paired over folds. Convergence is within 300 sweeps. Survival-active programs are counted in the fit to all training patients. The one-standard-error rule selects the smallest K whose mean C reaches the best mean minus its SE. Independently reviewed on November 12, 2026.

K	Mean C (SE)	Difference from K = 7 (SE)	Folds converged	Survival-active programs	Selected
2	0.611 (0.022)	−0.047 (0.009)	5/5	1	
3	0.617 (0.019)	−0.040 (0.010)	5/5	1	
4	0.620 (0.024)	−0.038 (0.009)	5/5	2	
5	0.655 (0.029)	−0.003 (0.007)	5/5	3	One-SE rule
6	0.659 (0.022)	+0.001 (0.005)	4/5	3	
7	0.658 (0.026)		3/5	3	
8	0.656 (0.027)	−0.001 (0.003)	3/5	3	
9	0.649 (0.022)	−0.009 (0.005)	2/5	4	
10	0.666 (0.027)	+0.008 (0.004)	2/5	4	
11	0.661 (0.021)	+0.003 (0.007)	1/5	4	
12	0.654 (0.020)	−0.004 (0.007)	0/5	4	

4.7 Rank and ELBO pruning

The pruning simulation asks whether ELBO pruning recovers the planted program and the true rank. For a flat coefficient likelihood the coefficient update returns the prior moments and an exact zero coefficient KL. Ordinary and protected pruning both lost the planted program in most datasets and retained too few programs (Table 10), so the fit retains all 7 starting columns.

Table 10: Pruning simulation. Converged counts fits whose relative change in the ELBO fell below the stopping tolerance. Outputs stable counts fits whose last-sweep change in the reconstruction and risk score was below 10^-4. The saved summary gives the count of planted-program recoveries and the median retained rank.

Arm	Noise	Recovered	Mean false-active	Converged	Outputs stable	Median rank
ordinary	0.25	2/20	1.10	20/20	0/20	2
protected	0.25	2/20	1.10	20/20	0/20	2
reference	0.25	1/20	1.15	20/20	0/20	7
unpruned	0.25	17/20	1.05	20/20	0/20	7
ordinary	1.00	2/20	0.20	20/20	3/20	1
protected	1.00	2/20	0.20	20/20	3/20	1
reference	1.00	1/20	0.40	20/20	0/20	7
unpruned	1.00	19/20	0.35	20/20	0/20	7

4.8 Comparison with other prognostic models

The supervised and two-step scores in this section are those of the supervised YFB fit ($w_S = 1$, 2,218 genes, rank 7) (Section 1.7). Every comparator was trained on the 273 training patients only, and each score's direction was fixed on the training data. DeSurv used the authors' code, preprocessing and published settings (rank 3), which the authors selected on the same training cohorts (Young et al. 2026). The elastic-net Cox model on the 2,218-gene set chose $\alpha = 0$ by cross-validation on the training folds. The = 0.75 and lasso paths reached glmnet's iteration limit at small penalties.

PurIST entered as a Cox model on its basal-like probability (Rashid et al. 2020). No clinical model was fitted, because no stage, grade, age or margin field is recorded in both training cohorts and all five external series.

Table 11: External performance of the fixed supervised fit and comparators trained on the same patients. Mean C is the equal-weight mean over the five series. The HR is per pooled external SD from a Cox model stratified by series, with a robust interval clustered on donor. The difference is supervised minus comparator C, averaged over series, with a 95% donor-bootstrap interval. Model: supervised YFB fit (w_S = 1, 2,218 genes, rank 7) and its two-step comparator. PurIST, elastic-net Cox and DeSurv are separate comparators trained on the same patients.

Model	Mean C	HR per SD (95% CI)	Difference (95% CI)
Supervised	0.620	1.37 (1.22 to 1.53)	
Two-step	0.613	1.35 (1.21 to 1.51)	+0.007 (−0.003 to 0.017)
PurIST	0.599	1.25 (1.13 to 1.38)	+0.021 (−0.013 to 0.057)
Elastic-net Cox	0.619	1.34 (1.20 to 1.49)	+0.001 (−0.015 to 0.018)
DeSurv	0.614	1.43 (1.25 to 1.64)	+0.007 (−0.017 to 0.030)

Every paired concordance interval included zero (Table 11). Unsupervised EBMF followed by Cox regression gave differences, supervised fit minus EBMF, of +0.007, +0.045, +0.040 at ranks 7, 15 and 40. At ranks 15 and 40 the unpenalized Cox step overfit, with external calibration slopes of 0.10 to 0.49, so those differences describe that pipeline rather than the information in a larger factorization.

4.8.1 Risk groups from training cutpoints

One cutpoint per score maximized the log-rank statistic on the out-of-fold training scores, with at least 20% of patients per group. It was applied unchanged in every external series. The cutpoints were −0.046 training SDs for the supervised score and 0.526 for the two-step score. The pooled high-versus-low HRs were 1.52 (1.24 to 1.87) and 1.69 (1.32 to 2.16). They are not comparable, because the supervised cutpoint places 297 of 616 external records in the high-risk group and the two-step cutpoint places 144. A smaller, more extreme group gives a larger HR at equal ranking ability. After adjusting for PurIST subtype the supervised score's pooled HR per training SD was 1.48, and after adjusting for DeCAF it was 1.40. Figure 6 displays the groups formed by the training cutpoints. The program section also gives within-series tertiles as a separate description of ranking. The out-of-fold scores used the feature matrix of the training fit, whose per-cohort scaling included held-out training-fold expression. The cutpoint search used the pooled out-of-fold training scores and their outcomes.

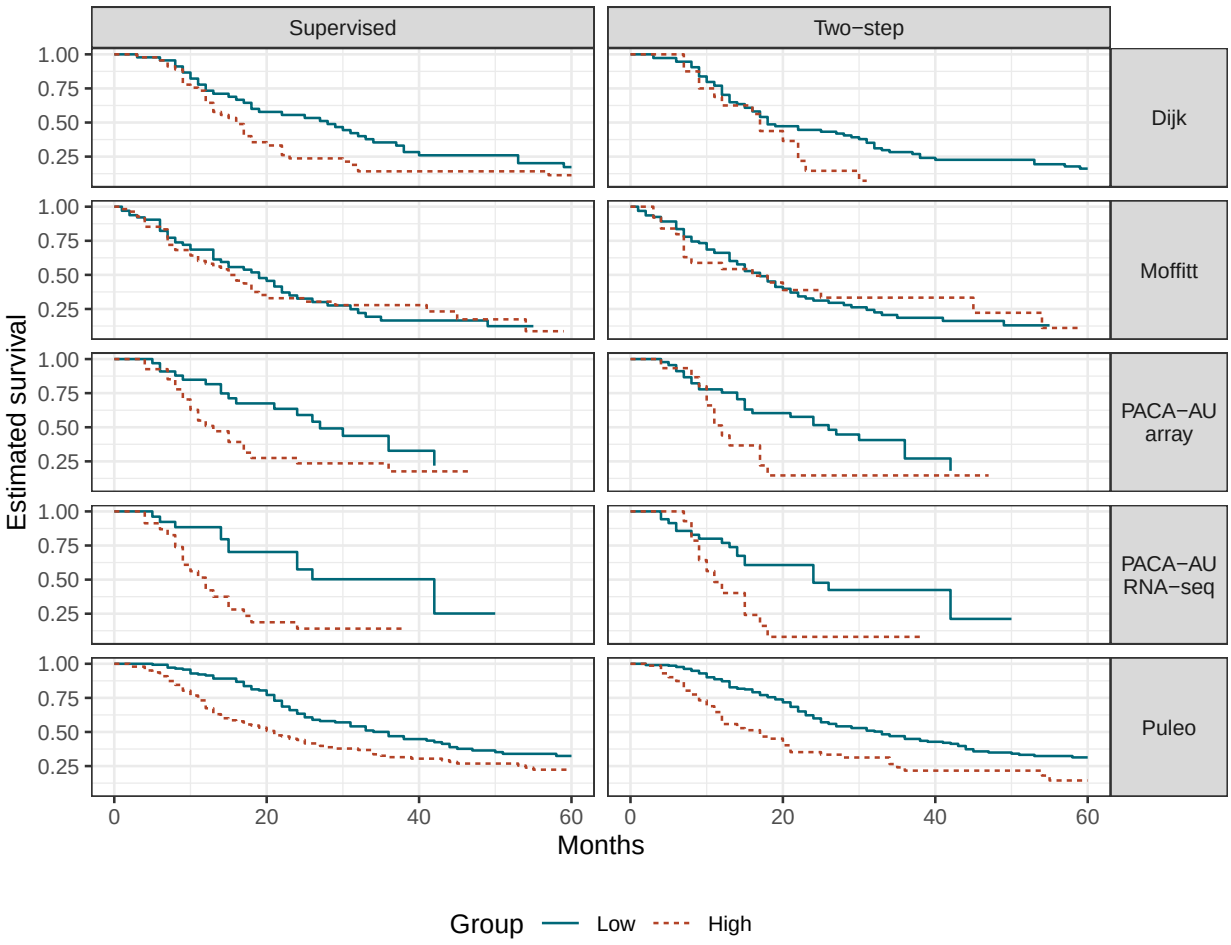


Figure 6: External Kaplan–Meier estimates by fixed training cutpoint. Each model uses its own cutpoint. Curves are drawn from saved monthly survival estimates. The numbers at risk are in Appendix C. Model: supervised YFB fit (w_S = 1, 2,218 genes, rank 7) and its two-step comparator.

These fixed groups and the subtype-adjusted associations are exploratory. Differences between high-versus-low hazard ratios also reflect differences in the patients assigned to each group.

4.8.2 Program stability in training folds

A main prognostic program was defined before the analysis by a standardized coefficient at least 0.25 of the largest, at least 5% of the signed risk-score variance and survival activity in at least four of the five training folds.

Table 12: Program classification rule on the supervised training fit. The ratio is | _k| SD(Y F_k) relative to the largest program. The signed share is Cov(_k,)/Var( ). Active folds count training folds in which a loading matched with absolute correlation at least 0.7 was survival-active. The precision share is an upper bound on survival's median share of the loading precision.

Program	Active	Ratio	Signed share	Active folds	Main program	Precision share
2	Yes	1.00	0.632	5	Yes	7.9e-07
3	Yes	0.54	0.179	4	Yes	6.3e-07
4	No	0.38	−0.010	0	No	5.4e-07
5	Yes	0.57	0.202	3	No	2.1e-06
7	No	0.12	−0.003	1	No	3.4e-07

Programs 2 and 3 met the rule. Program 5 met the coefficient and share thresholds but was active in only 3 folds. In the saved fits of the cross-validated-weight simulation the coefficient-and-share part kept the planted program in 80 of 80 fits and rejected 35 of 36 false-active directions. In the two-step fits of the near-threshold simulation it rejected 61 of 128. The fold-stability part could not be checked in simulation.

4.8.3 Mixed-signal simulation

A simulation planted one strong and one near-threshold prognostic program with equal standardized effects at 2,000 genes.

Table 13: Mixed-signal simulation, ten datasets per arm. All fits converged and every arm recovered the strong program in every dataset. Precision is the fraction of the 50 genes with the largest absolute fitted loading on the matched program that lie in the planted support. The mean includes datasets whose best match has absolute correlation below 0.7.

Arm	Near-threshold recovered	Top-50 precision	Validation C
Two-step	9/10	0.826	0.731
Supervised, w_S = 1	9/10	0.858	0.735
Supervised, joint , w_S = 1	9/10	0.860	0.734
Supervised, w_S = 5	2/10	0.346	0.667

At $w_S = 1$ the supervised and two-step fits recovered the near-threshold program equally often. A fixed $w_S = 5$ lost it in most datasets and had lower validation C.

4.8.4 Recovery across simulation designs

The matched two-step fit recovered the planted program in the representative cell of the cross-validated-weight simulation. The largest unit-weight recovery difference in the near-threshold simulation favoured the supervised fit and occurred at the smallest planted share with strong survival (Table 6). In the mixed-signal simulation, the two unit-weight fits had the same near-threshold recovery as the two-step fit. Figure 7 shows the accompanying differences in false activity, convergence and calibration. On PDAC data the loading agreement and the small Cox contribution to loading precision are consistent with an effect through the coefficient fit (Section 4.2).

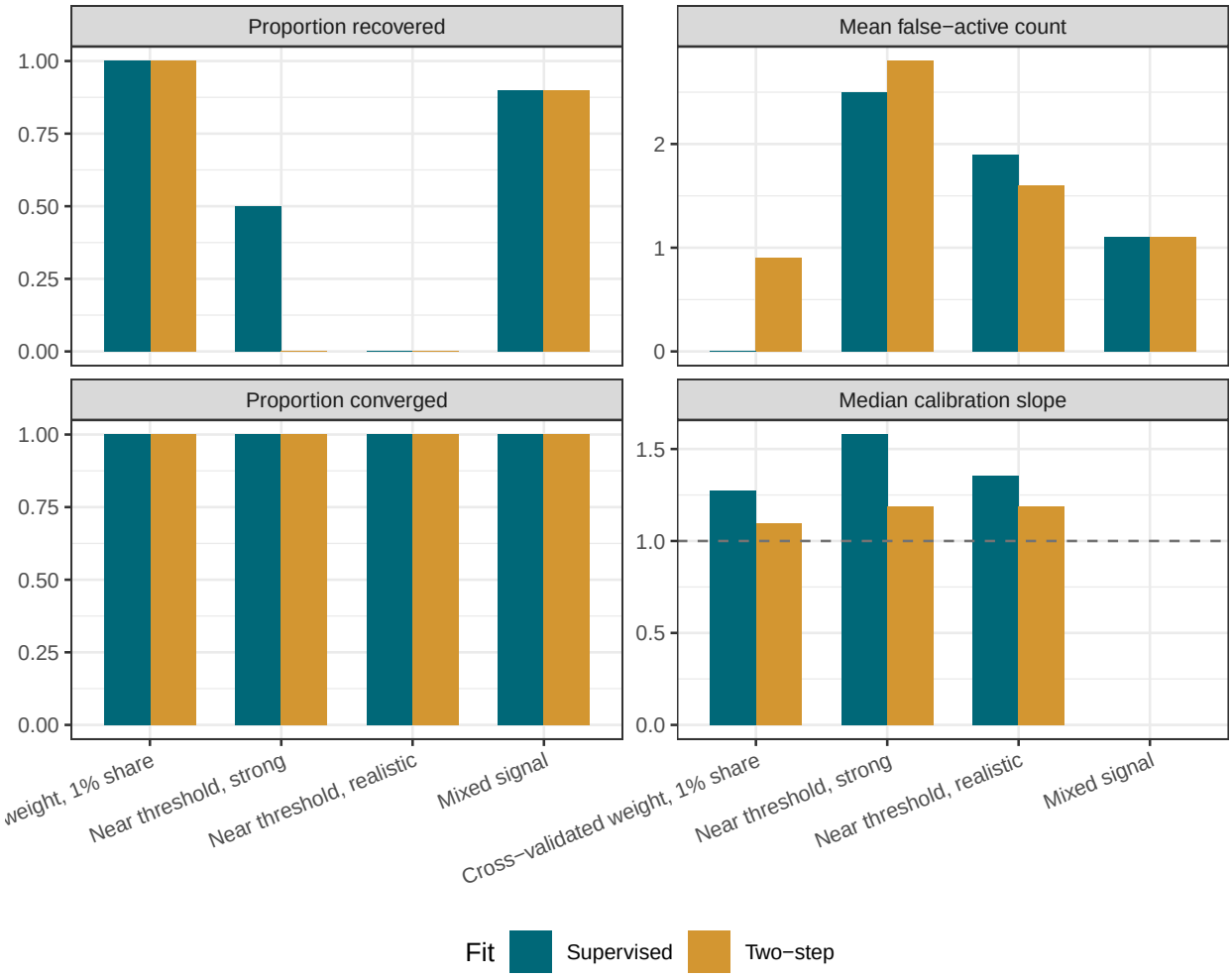


Figure 7: Matched simulation results. The cross-validated-weight simulation panel shows the 1% planted-variance cell with the cross-validation-selected weight in the block-weighted fit. The near-threshold simulation panel shows the smallest planted share at both survival strengths. The mixed-signal cell reports recovery of the weaker loading and uses one row per fit for false activity and convergence. Calibration slopes were not saved for the mixed-signal experiment.

5 Programs of the supervised fit

The programs in this section are those of the supervised YFB fit ($w_S = 1$, 2,218 genes, rank 7) (Section 1.7), and every external association reported here comes from that model. The joint fit retained 7 programs. Of those, 3 were survival-active (Table 14). Program 2 had a protective coefficient at its positive pole, where the leading genes included classical epithelial and exocrine markers. Its negative pole included hypoxia and glycolysis genes. Program 3 had an adverse coefficient at its ribosomal and proliferative pole. Program 5 also had an adverse coefficient, with enrichment of mitotic and stromal-immune signatures.

Table 14: Programs of the supervised training fit. Coefficients are log hazard ratios per SD of the program score, with posterior SDs in parentheses. The positive and negative poles list the genes with the largest signed loadings in each direction.

Program	Coefficient (SD)	Active	Positive pole	Negative pole
1	0.000 (0.000)	no	TCF4, FBN1, BNC2, GAS7	RAB25, PKP3, GALE, RASSF7
2	−0.405 (0.084)	yes	ATP2A3, ACACB, METTL7A, ADH1B	LOXL2, MMP14, P4HA1, LDHA
3	0.218 (0.073)	yes	RPS7, RPL6, NCAPD2, EEF1D	KIAA1211, SYTL2, CTSE, TSPAN8
4	0.154 (0.082)	no	CTRB2, CTRB1, CPB1, REG1B	BANK1, STK17B, EMB, MS4A1
5	0.230 (0.082)	yes	UHMK1, ARHGAP5, TAOK1, RAD21	CHPF, RCN3, MXRA8, LOXL1
6	0.000 (0.000)	no	IFI30, LILRB4, FCER1G, CD74	RAP1GAP2, IGF1R, ARHGAP32, PBX1
7	−0.048 (0.055)	no	ANO1, BCL9L, PPL, VSTM2L	PRDX4, PPIB, NUCB2, P4HB

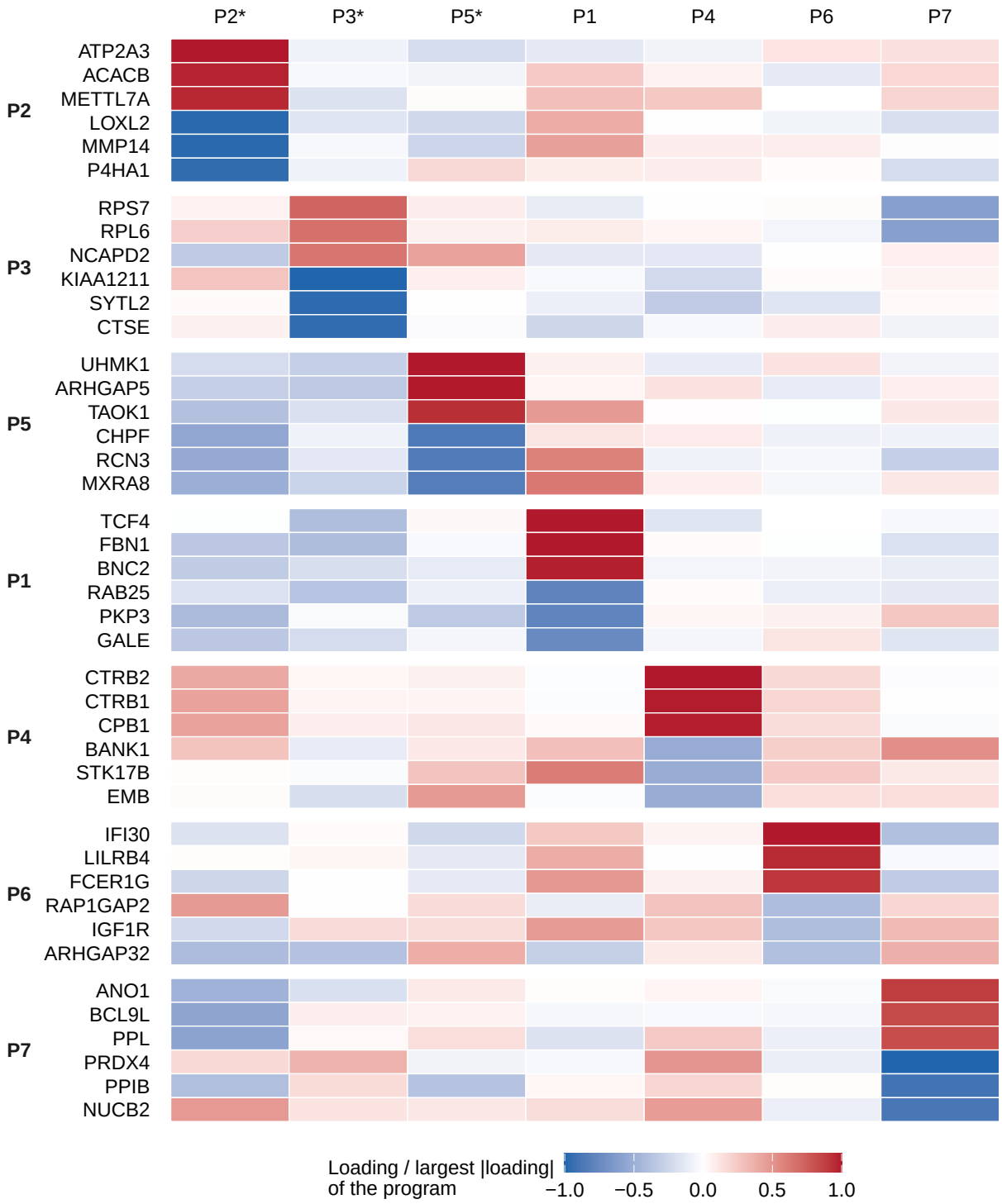


Figure 8: Scaled loadings of selected genes across the fitted programs. Asterisks indicate survival-active programs.

5.1 Enrichment and published signatures

Enrichment was calculated on signed loadings among the modeled genes. The leading sets are in Table 15. Program 3's influenza-labeled sets share leading ribosomal genes in the saved enrichment results.

Table 15: Leading enrichment at each pole of the survival-active programs. NES denotes normalized enrichment score. The p-values are adjusted within program.

Program	Collection	Gene set	NES	Adjusted p
2	PDAC	DeSurv D1 ClassicalTumor	2.83	9.91e-21
2	Hallmark	EPITHELIAL MESENCHYMAL TRANSITION	−2.82	1.54e-17
3	Reactome	INFLUENZA INFECTION	4.64	4.35e-44
3	PDAC	Moffitt Classical	−2.28	3.22e-10
5	PDAC	DeSurv D2 StromalImmune	2.85	3.80e-30
5	Hallmark	EPITHELIAL MESENCHYMAL TRANSITION	−2.06	3.47e-06

5 Reactome p-values were unestimable and are retained as missing in the source CSV. Spearman correlations with published signatures were computed within series using the available standardized expression genes. In the pooled training data, Programs 2 and 3 were inversely associated with the Moffitt basal-like and classical signatures, respectively. The corresponding external ranges are in Table 16. Correlations between Program 5 and the stromal-immune signature were weaker. These descriptive associations characterize fixed programs.

Table 16: Spearman correlations with selected subtype and stromal signatures. The external interval is the range of point estimates across series. PurIST uses the saved basal-like probabilities. Model: supervised YFB fit (w_S = 1, 2,218 genes, rank 7).

Program	Signature	Training	External	External PurIST
2	Moffitt basal-like	−0.71	−0.78 to −0.71	−0.64 to −0.45
3	Moffitt classical	−0.76	−0.84 to −0.56	0.30 to 0.79
5	DeSurv stromal-immune	0.21	−0.03 to 0.23	−0.25 to −0.13

The loadings of the supervised fit and of the $w_S = 0$ fit agreed at $|r| = 1.000$ for every matched program (Section 4.2).

5.2 Evidence for the individual programs

Program 2 combines the clearest external direction agreement with survival activity in 5 training folds. Its protective pole is associated with classical epithelial and exocrine expression. Its adverse pole includes hypoxia and glycolysis genes. The separate external Cox estimates agree in direction with its training coefficient across all scored series.

Program 3 was active in 4 training folds. Its adverse pole includes ribosomal and proliferative genes, and its opposite pole correlates with the classical signature. The unadjusted external estimates generally follow the training direction (Table 17).

Program 5 had an adverse full-fit coefficient and enrichment for mitotic and stromal-immune signatures. It was active in 3 training folds. Its pooled external HR per training SD was 0.87

(donor-clustered 95% CI 0.76 to 1.00), opposite in direction to its training coefficient, and 3 of the 5 series had HRs below one. Its biological interpretation is exploratory. The retention rule in Section 4.8.2 reflects that instability.

5.3 External program associations and risk groups

Separate external Cox regressions give descriptive associations per series SD of each active program score. The numbers of estimates in the training direction, in program order, were 5, 4, 2 of 5. These estimates are marginal. The training coefficients condition on the other program scores.

Table 17: External program associations from separate Cox regressions. Entries are hazard ratios and Wald intervals per SD of the program score within series. Model: supervised YFB fit (w_S = 1, 2,218 genes, rank 7).

Series	Program 2	Program 3	Program 5
Dijk	0.64 (0.49 to 0.83)	0.98 (0.79 to 1.21)	0.77 (0.62 to 0.96)
Moffitt	0.90 (0.73 to 1.12)	1.12 (0.90 to 1.40)	0.87 (0.69 to 1.09)
PACA-AU array	0.78 (0.58 to 1.06)	1.28 (0.98 to 1.68)	1.08 (0.79 to 1.47)
PACA-AU RNA-seq	0.71 (0.51 to 0.99)	1.84 (1.31 to 2.59)	1.21 (0.82 to 1.79)
Puleo	0.72 (0.61 to 0.85)	1.37 (1.18 to 1.59)	0.93 (0.81 to 1.07)

Kaplan–Meier curves use within-series tertiles of the fixed risk score or fixed program scores of the supervised fit. Program scores are oriented using the training coefficients. The tertiles describe relative ranking within each series. They use a different set of cutpoints in each series.

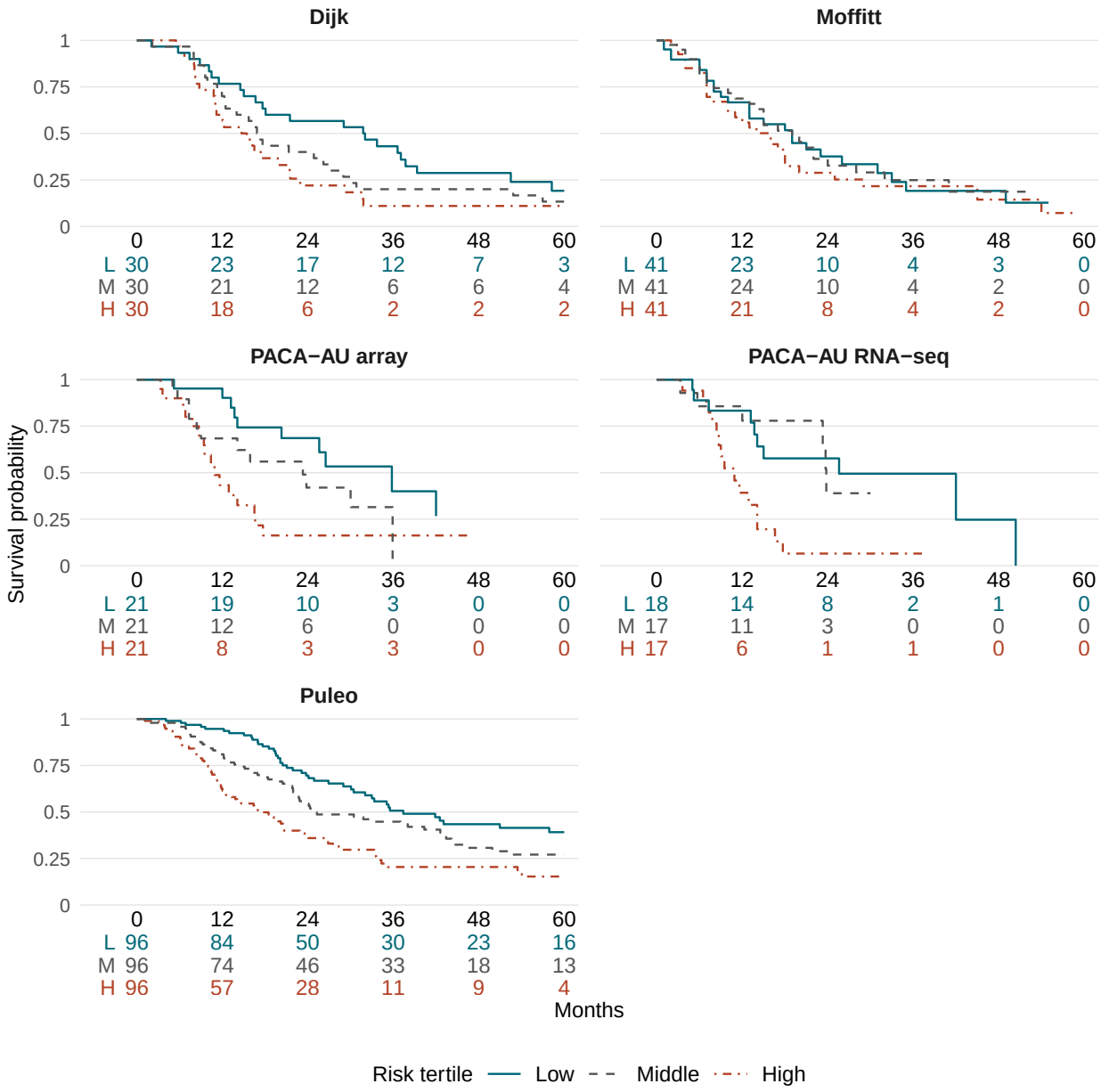


Figure 9: External Kaplan–Meier curves for tertiles of the fixed supervised risk score, with numbers at risk. Model: supervised YFB fit (w_S = 1, 2,218 genes, rank 7).

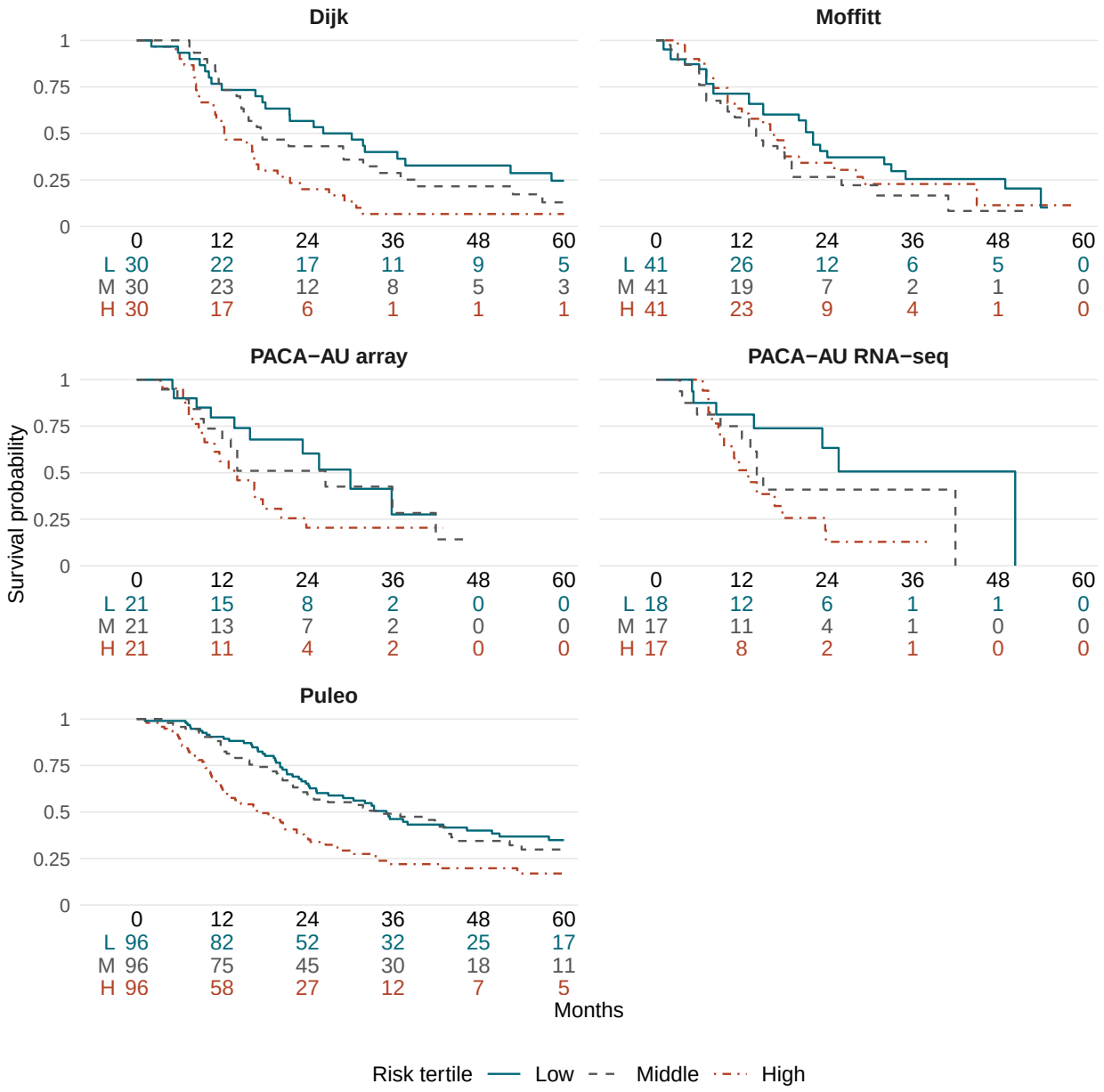


Figure 10: External Kaplan–Meier curves by tertile of the Program 2 score oriented in the adverse training direction, with numbers at risk. Model: supervised YFB fit (w_S = 1, 2,218 genes, rank 7).

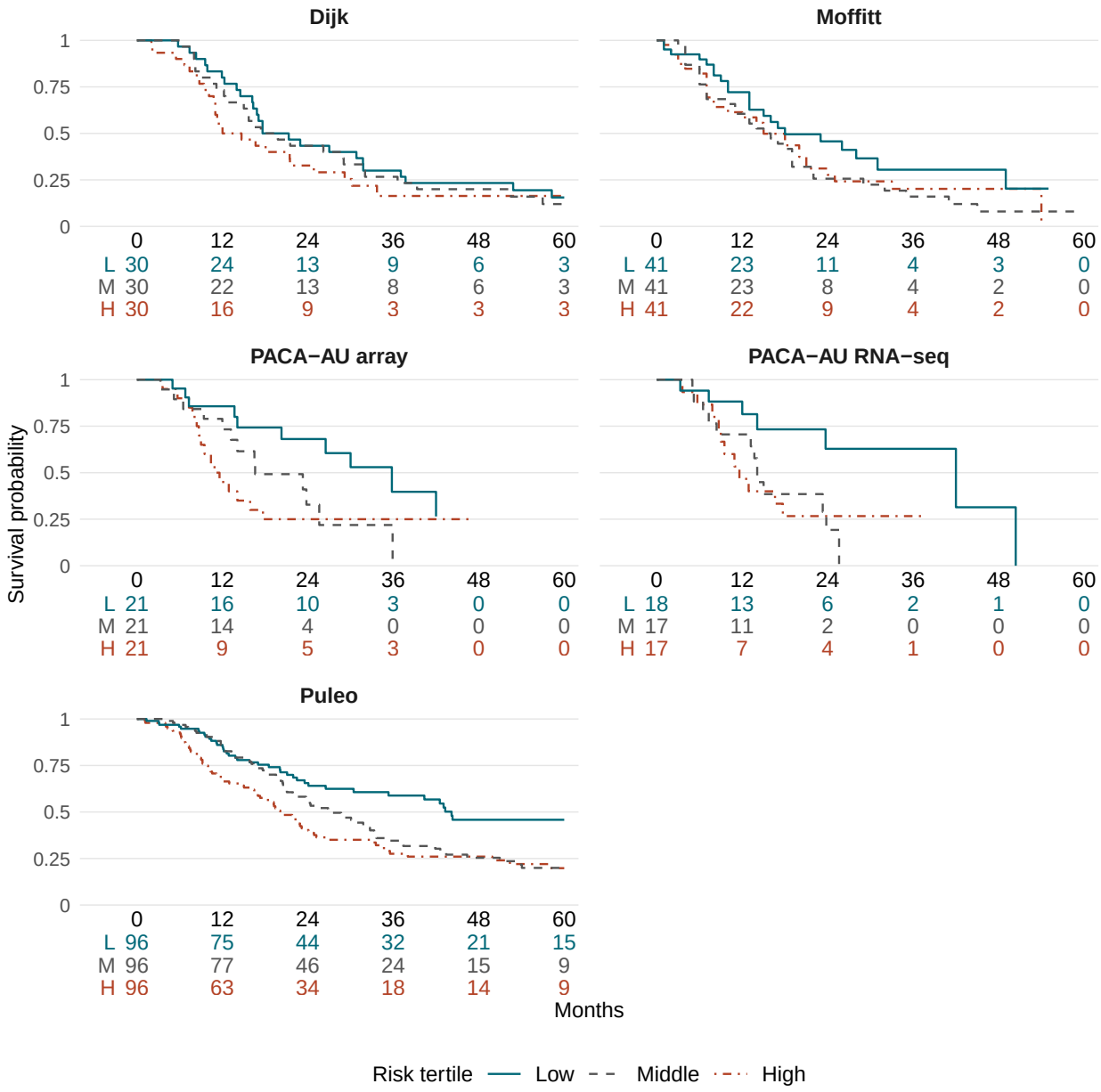


Figure 11: External Kaplan–Meier curves by tertile of the Program 3 score oriented in the adverse training direction, with numbers at risk. Model: supervised YFB fit (w_S = 1, 2,218 genes, rank 7).

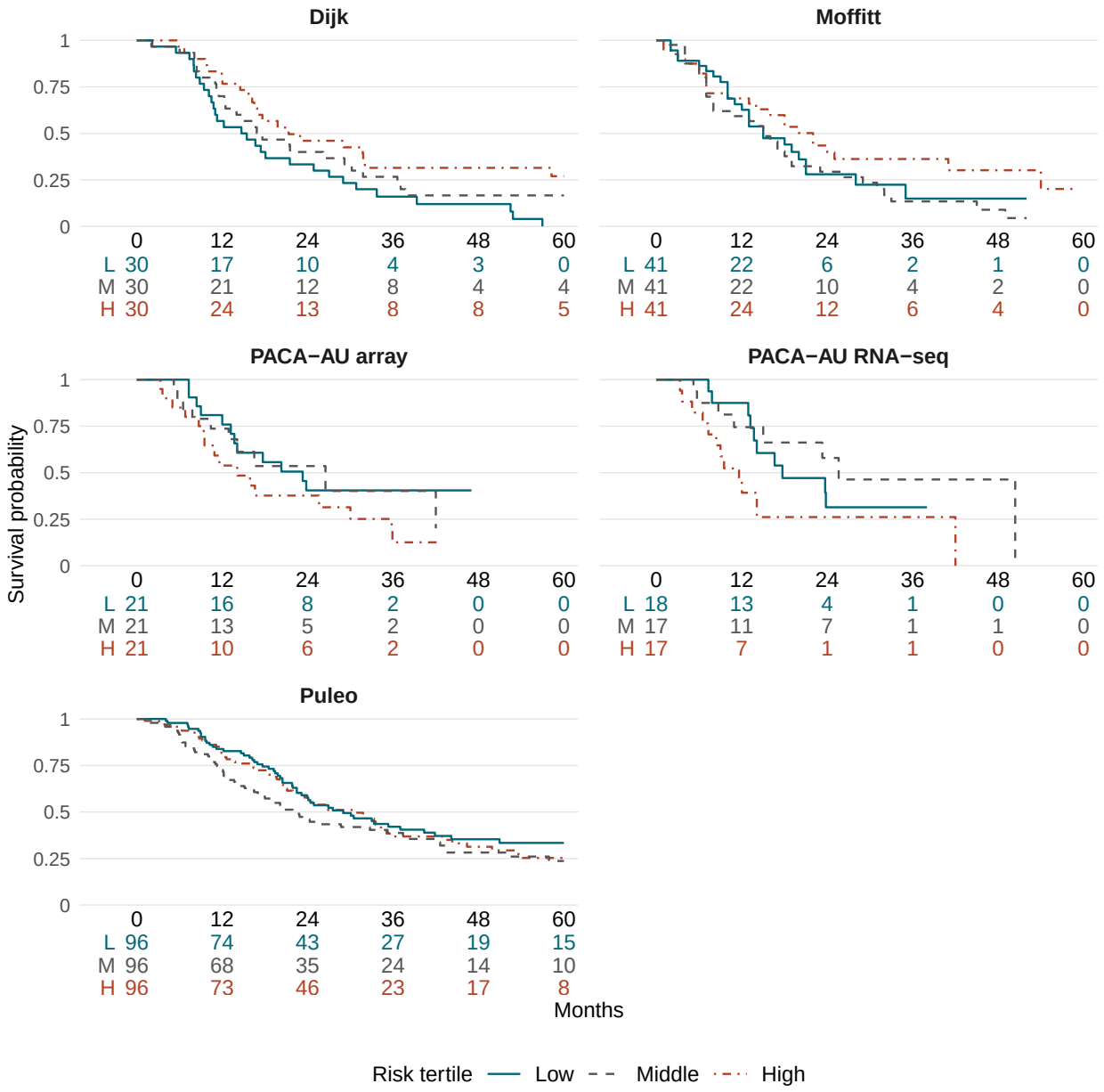


Figure 12: External Kaplan–Meier curves by tertile of the Program 5 score oriented in the adverse training direction, with numbers at risk. Model: supervised YFB fit (w_S = 1, 2,218 genes, rank 7).

6 Discussion

6.1 Interpretation

For the supervised YFB fit ($w_S = 1$, 2,218 genes, rank 7), the external concordance difference was small and its interval included zero. The estimated loadings correlated 1.000 with the unsupervised loadings at the unit survival weight. DeSurv and elastic-net Cox trained on the same patients performed similarly. The joint coefficient fit accounts for posterior loading uncertainty and uses a

shrinkage prior. Calibration slopes were closer to one in most series.

In the simulation cells with consistent recovery, the two-step fit already recovered the prognostic program. The near-threshold simulations isolated a setting where supervision could change discovery. Recovery and concordance improved when survival was strong, while the differences were smaller at realistic survival strength.

6.2 Limitations

The gene screen, the input scaling and the rank of the supervised YFB fit ($w_S = 1$, 2,218 genes, rank 7) were chosen with external concordance before the training-only rule, and every external series informed at least one of the choices recorded in the audit (Section 2.2). Training cross-validation now supports the screen and the scaling and places rank 7 on the flat part of its concordance curve (Section 4.6). Because the external series informed these choices first, the external results still describe a model whose setup those cohorts helped choose. The training comparison used one run of five-fold cross-validation, that is, one random partition of the patients into folds. Close comparisons are repeated with more random partitions, and the plateau from $K = 5$ to 12 and the screen comparison are that case. The repetition with four more random partitions was running on November 12. The PACA-AU array and RNA-seq records both contribute to the pooled estimate. The donor bootstrap and the donor-clustered HR intervals account for their shared donors while retaining both series in the estimand. DeSurv used its published settings rather than a new search on our training folds. 2 of 5 supervised training folds stopped at the sweep cap. In simulation the coefficients at $w_S = 1$ are attenuated at the larger gene count. These studies use a common simulation generator.

6.3 Continuing work

Proposed supervision weights include a power on the Cox partial likelihood chosen to match the information in a correctly specified model, downweighting expression and normalizing the likelihood terms. Each would be compared in simulation, on recovery of the planted program, calibration slopes, convergence and the risk-score contribution of programs that match no planted program, before application to the real data.

The analyses of Section 3 compare priors on the loadings and the coefficients, fit methylation alone, test a fitter for several data types with shared activities and separate loading priors, and compare the model with the $\eta = L\beta$ parameterization. The working model now uses rank 5, selected by the one-standard-error rule (Section 4.6). Once the results of Section 3 are reviewed, the working model is fixed on training data and scored once on the external cohorts (Section 3.7). Every later choice follows the training-only rule of Section 2.2.

Appendix A. Model derivation

The canonical single-modality procedure uses training expression Y and survival times and indicators. It returns posterior moments of L, F and β, gene precisions and the tracked variational objective.

1. Center and scale the input according to the training preprocessing recipe. Initialize the factorization using a truncated singular value decomposition of the resulting Y.
2. Compute the posterior mean and variance of η. Re-expand the Cox partial log-likelihood about the current mean predictor, using the diagonal curvature to obtain a working normal-means objective.
3. For each program, update β_k using the first and second moments of YF_k. Update L_k from the expression residual. Update the entries of F_k using both likelihood contributions and refresh the projected scores within the loading update. Use empirical Bayes normal-means fits for the prior parameters.
4. Update the gene-specific noise precisions. Record the monitored objective including the weighted Cox term and the Kullback–Leibler terms. Rescale nonzero programs to unit training projection SD and check the relative change in the objective.
5. In each external series, apply the fixed projection loadings and coefficients to the available training genes. Harrell's concordance uses the direction learned from training survival.

The full derivation gives the update formulas, the balance of expression and survival information and the numerical checks. The companion `verify_derivation.R` evaluates individual updates against the expected objective.

Appendix B. Changes since the November 6 version

- **External cohorts.** The external cohorts only score fixed models. Every modeling choice is made by simulation and training cross-validation (Section 2.2).
- **Model names.** The model behind the external results is named by its configuration, and every external table and figure names it in the caption (Section 1.7).
- **Historical comparisons removed.** This version compares the current model only with the rank-matched two-step fit, EBMF at ranks 15 and 40 followed by Cox, DeSurv, elastic-net Cox and PurIST. Removed: the pre-registered October 15 supervision check, the $\alpha_F = 0$ benchmark and its single-log refit, the 40-factor LASSO comparator, the matches with historical loadings, the table of past external-cohort uses and the earlier multimodal appendix.
- **Work of November 9–13.** Section 3 describes the week's analyses with placeholders for their results.
- **Gene screen, input scaling and rank.** New subsections in the Methods (Section 1.6) and the Results (Section 4.6) report the training cross-validation of these three choices. It keeps the screen and the scaling and selects rank 5 for the working model. Its result replaces the four placeholders of Section 3.6, and the Summary, Limitations and Continuing work describe it.
- **Prior sensitivity.** The comparisons of priors on the scores and loadings and on the survival coefficients are complete. Their results replace two placeholders in Section 3, and Appendix D gives the tables.

No number retained from the November 6 version has changed.

Appendix C. Comparator and risk-group details

Table 18: External discrimination and calibration by series. Concordance intervals are the saved series-level bootstrap intervals. Calibration is the Cox slope with its standard error. The paired donor-level comparison appears in the main text. Model: supervised YFB fit (w_S = 1, 2,218 genes, rank 7) and its two-step comparator. PurIST, elastic-net Cox and DeSurv are separate comparators trained on the same patients.

Series	Model	Records	Events	C (95% CI)	Slope (SE)
Dijk	Supervised	90	81	0.584 (0.513 to 0.658)	0.55 (0.36)
Dijk	Two-step	90	81	0.567 (0.497 to 0.640)	0.35 (0.31)
Dijk	PurIST	90	81	0.640 (0.574 to 0.706)	0.98 (0.29)
Dijk	Elastic-net Cox	90	81	0.616 (0.545 to 0.685)	0.84 (0.36)
Dijk	DeSurv	90	81	0.604 (0.532 to 0.679)	0.45 (0.18)
Moffitt	Supervised	123	83	0.537 (0.459 to 0.611)	0.36 (0.36)
Moffitt	Two-step	123	83	0.531 (0.449 to 0.604)	0.28 (0.32)
Moffitt	PurIST	123	83	0.569 (0.500 to 0.637)	0.37 (0.26)
Moffitt	Elastic-net Cox	123	83	0.562 (0.486 to 0.636)	0.49 (0.39)
Moffitt	DeSurv	123	83	0.558 (0.480 to 0.634)	0.46 (0.26)
PACA-AU array	Supervised	63	38	0.656 (0.562 to 0.751)	0.78 (0.33)
PACA-AU array	Two-step	63	38	0.656 (0.558 to 0.751)	0.73 (0.30)
PACA-AU array	PurIST	63	38	0.579 (0.467 to 0.685)	0.20 (0.56)
PACA-AU array	Elastic-net Cox	63	38	0.645 (0.545 to 0.740)	0.77 (0.38)
PACA-AU array	DeSurv	63	38	0.642 (0.541 to 0.742)	0.63 (0.27)
PACA-AU RNA-seq	Supervised	52	31	0.681 (0.579 to 0.780)	1.57 (0.43)
PACA-AU RNA-seq	Two-step	52	31	0.671 (0.574 to 0.771)	1.26 (0.37)
PACA-AU RNA-seq	PurIST	52	31	0.607 (0.502 to 0.718)	0.76 (0.41)
PACA-AU RNA-seq	Elastic-net Cox	52	31	0.634 (0.536 to 0.736)	1.29 (0.45)
PACA-AU RNA-seq	DeSurv	52	31	0.630 (0.516 to 0.741)	0.67 (0.25)
Puleo	Supervised	288	181	0.644 (0.599 to 0.686)	0.99 (0.20)
Puleo	Two-step	288	181	0.643 (0.599 to 0.686)	0.80 (0.17)
Puleo	PurIST	288	181	0.602 (0.551 to 0.649)	0.58 (0.19)
Puleo	Elastic-net Cox	288	181	0.638 (0.595 to 0.681)	0.97 (0.21)
Puleo	DeSurv	288	181	0.633 (0.588 to 0.676)	0.58 (0.14)

Table 19: Numbers at risk for the fixed-cutpoint curves. Entries are low-risk / high-risk counts by time in months. Model: supervised YFB fit (w_S = 1, 2,218 genes, rank 7) and its two-step comparator.

Model	Series	0	12	24	36	48	60
Supervised	Dijk	45 / 45	33 / 29	25 / 10	15 / 5	10 / 5	5 / 4
Two-step	Dijk	74 / 16	52 / 10	33 / 2	20 / 0	15 / 0	9 / 0
Supervised	Moffitt	67 / 56	38 / 30	15 / 13	6 / 6	4 / 3	0 / 0
Two-step	Moffitt	97 / 26	55 / 13	21 / 7	8 / 4	5 / 2	0 / 0
Supervised	PACA-AU array	35 / 28	26 / 13	13 / 6	3 / 3	0 / 0	0 / 0
Two-step	PACA-AU array	47 / 16	33 / 6	17 / 2	4 / 2	0 / 0	0 / 0

Table 19: Numbers at risk for the fixed-cutpoint curves. Entries are low-risk / high-risk counts by time in months. Model: supervised YFB fit (w_S = 1, 2,218 genes, rank 7) and its two-step comparator.

Model	Series	0	12	24	36	48	60
Supervised	PACA-AU RNA-seq	28 / 24	21 / 10	9 / 3	2 / 1	1 / 0	0 / 0
Two-step	PACA-AU RNA-seq	38 / 14	26 / 5	11 / 1	2 / 1	1 / 0	0 / 0
Supervised	Puleo	144 / 144	122 / 93	72 / 52	44 / 30	30 / 20	20 / 13
Two-step	Puleo	216 / 72	177 / 38	105 / 19	65 / 9	43 / 7	29 / 4

Appendix D. Sensitivity to the priors

These two comparisons were completed and independently reviewed on November 12, 2026. Both use rank 7, the rank of the reported fit, and are comparisons between priors, not tests of each prior against an absolute standard. Their full results are in `docs/findings/2026-11-12-lf-prior-comparison-findings.md`, `docs/findings/2026-11-12-lf-prior-followup-findings.md` and `docs/findings/2026-11-12-beta-prior-comparison-findings.md`.

Priors on the scores and loadings

The current model uses point-Laplace priors on L and F with expression z-scored within each cohort. The alternative is a point-exponential prior, which keeps every score and loading nonnegative, as in a nonnegative matrix factorization. It needs nonnegative input, so expression is divided by each gene's within-cohort standard deviation without centering, and the Cox projection is centered on the training data. Everything else is the current model's: $w_S = 1$, a Normal coefficient prior, $K = 7$ without pruning, at most 300 sweeps and tolerance 10^{-5}. The simulation uses the three variance cells of the fixed-weight simulation at 2,000 genes and the strong-survival cell of the near-threshold simulation at a planted share of 0.05%, with 10 datasets each. The planted scores and loadings are exponential, so the truth favors neither prior. The training comparison uses the five folds of Section 4.1 on the pooled 273 patients and counts only pairs of patients from the same cohort.

In simulation the point-exponential prior recovered the planted program as often as point-Laplace, and more often near the detection threshold (8 against 5 of 10), and every fit converged (Table 20). The point-exponential prior also marked more programs survival-active that matched no planted program, with means of 2.6 to 4.3 per fit against 0.0 to 3.0 for point-Laplace. Such programs are expected, and in the shifted-input fits, where their size was measured, their coefficients were mostly a tenth to a quarter of the planted program's.

Table 20: Point-Laplace and point-exponential priors on the scores and loadings in simulation at 2,000 genes. Entries give point-Laplace first and point-exponential second. Recovered and converged are counts of datasets. False-active is the mean number of false-active programs per fit. Validation C and the calibration slope are medians over datasets. The 0.05% cell is the strong-survival cell of the near-threshold simulation. Independently reviewed on November 12, 2026.

Planted share	Recovered	False-active programs	Converged	Validation C	Calibration slope
0.05%	5 / 8	3.0 / 2.6	10 / 10	0.832 / 0.839	1.58 / 1.26
0.25%	10 / 10	0.9 / 3.0	10 / 10	0.947 / 0.940	1.45 / 1.39
1%	10 / 10	0.1 / 4.0	10 / 10	0.974 / 0.971	2.16 / 1.82
5%	10 / 10	0.0 / 4.3	10 / 10	0.987 / 0.982	3.23 / 2.07

On the training data the point-exponential fit lost concordance and described the difference between the cohorts (Table 21). Dividing by the within-cohort standard deviation keeps each cohort's mean level. The median gene mean was 11.0 in TCGA-PAAD and 15.5 in CPTAC, against a within-cohort standard deviation of 1. In every fold one program's scores correlated at least 0.998 with cohort membership, and no program was survival-active. Three further setups removed or absorbed that difference. Cohort strata center the projection and the baseline hazard within each cohort. A shifted z-score adds to each gene's z-scores a constant equal to minus their minimum over the

273 patients, so the input is nonnegative and both cohorts have the same gene means. The shift constants use all training patients, held-out folds included, and they enter the reconstruction, so a refitted model can change with them. The within-patient rank transform used by DeSurv ranks the genes within each patient and divides each gene's ranks by their within-cohort standard deviation. Each setup recovered part of the lost concordance, but all stayed below point-Laplace by more than one standard error, and no setup converged in more than 1 of 5 folds within 300 sweeps. The shifted z-score removed the association with cohort, but in simulation it behaved like the unshifted point-exponential fit, with 2.8 to 4.4 false-active programs per fit. Under strata the programs still correlate 0.998 with cohort membership, so they still separate the two cohorts, and the strata absorb that offset only in the survival model. The programs keep the loading directions of the unstratified fit, with larger column norms. The point-Laplace prior is kept, because it predicted better on the training data. At rank 5, the rank selected in Section 4.6, the point-exponential prior on expression divided by its within-cohort standard deviation gave within-cohort concordance 0.519 against 0.655 for point-Laplace.

Table 21: Priors on the scores and loadings in five-fold training cross-validation. Concordance counts only pairs of patients from the same cohort. The difference from the current model is paired over folds, with its standard error. Convergence is within 300 sweeps. Survival-active programs are the range over folds. Cohort correlation is the largest absolute correlation of a program's training scores with cohort membership over folds. Under strata the cohort offset is absorbed by the stratified baseline hazard. Independently reviewed on November 12, 2026.

Prior and input	Mean C	Difference (SE)	Folds converged	Survival-active programs	Cohort correlation
Point-Laplace, z-scores within cohort (current model)	0.658		3/5	2 to 4	0.05
Point-exponential, divided by within-cohort SD	0.573	−0.085 (0.031)	0/5	0	0.998
Point-exponential, divided by within-cohort SD, cohort strata	0.628	−0.030 (0.015)	0/5	3 to 4	0.998
Point-exponential, shifted z-scores	0.635	−0.023 (0.008)	1/5	2 to 4	0.05
Point-exponential, within-patient ranks	0.615	−0.043 (0.016)	0/5	2	1.000

Prior on the survival coefficients

The current model gives each coefficient its own Normal prior. The alternatives are a point-Laplace and a point-normal prior, either of which can set a coefficient exactly to zero. A prior with a point mass cannot be estimated from one coefficient, so both are shared across the seven programs. A shared Normal prior separates the effect of sharing from that of sparsity. All other settings, the simulation cells and the folds are those of the comparison above, with point-Laplace priors on the scores and loadings.

The four priors gave nearly the same fits (Table 22). The sparse priors lowered false activity slightly where the planted program was clear, to 0.3 against 0.9 at a share of 0.25%. Median calibration slopes differed by at most 0.03 between priors outside the 5% cell. In training cross-validation the shared point-Laplace prior lowered concordance by 0.008 (SE 0.002), and the shared point-normal prior by 0.002 (SE 0.004). Under the shared point-normal prior, the posterior probability that a

coefficient is nonzero was between 0.94 and 1.00 for all seven programs in the full training fit, so it did not single out the 3 survival-active programs. In simulation the mean number of programs per fit with an inclusion probability above 0.5 was 1.3 at a share of 0.25% and 1.0 at the larger shares, so the prior picked out the planted program alone when that program was clear. Near the threshold it was 4.2. The Normal prior is kept.

Table 22: Coefficient priors in simulation at 2,000 genes and in five-fold training cross-validation. Point-Laplace priors on the scores and loadings throughout. Simulation entries give the planted programs recovered and the mean number of false-active programs per fit. Every simulated fit converged. Training concordance counts pairs within cohort, and the difference from the Normal prior is paired over folds. Independently reviewed on November 12, 2026.

Measure	Normal (current model)	Shared point-Laplace	Shared point-normal	Shared Normal
Simulation, 0.05% share: recovered, false-active	5, 3.0	5, 3.3	4, 3.5	4, 4.1
Simulation, 0.25% share: recovered, false-active	10, 0.9	10, 0.3	10, 0.3	10, 1.5
Simulation, 1% share: recovered, false-active	10, 0.1	10, 0.0	10, 0.0	10, 0.4
Simulation, 5% share: recovered, false-active	10, 0.0	10, 0.0	10, 0.0	10, 0.0
Training CV: mean C	0.658	0.649	0.656	0.651
Difference from Normal (SE)		−0.008 (0.002)	−0.002 (0.004)	−0.007 (0.004)
Survival-active programs per fold	2.6	2.4	2.4	2.6
Folds converged	3/5	3/5	3/5	3/5

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